

GENERAL SPECIFICATION G4.2

Pipework

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G4.2 PIPEWORK

G4.2.1 General

- (a) The term pipework shall include all necessary supports, saddles, slings, fixing bolts and foundation bolts required to support the pipes and associated equipment.
- (b) Pipeline Expansion Systems: Design, supply and install pipeline expansion and anchorage systems on non-ambient temperature pipe systems and non-metallic pipe systems.
- (c) Pipework in buildings shall be arranged and designed so as to facilitate the erection, painting in situ and dismantling of any section for maintenance and to give a constant and uniform flow of fluid with minimum head loss. Where steel pipework is used, the number of flanges shall be kept to a minimum and the length of each unit of pipework take into consideration the ease of handling, installation and general appearance of the completed pipe system. Positions of flanges shall take into account any necessary concrete pipe supports or thrust blocks. Flanges shall not be cast into floors or walls and shall be located such that they can be readily dismantled.
- (d) Flexible joints/mechanical couplings shall be provided where necessary to facilitate dismantling and removal of plant and equipment easily without dismantling any pipework or supports and/or to allow for differential settlement of building structure. Flange adapter may only be omitted with the approval of the S.O..
- (e) Flexible joints/mechanical couplings shall be provided with tie bolts along the pipework unless otherwise specified or instructed by S.O.. Pipe supports shall not be used to absorb thrust.
- (f) All dismantling joints shall be tied through suitable welded flange or brackets directly welded to the pipework such that hydraulic thrust will not be transmitted to pipework supports.
- (g) Professional Engineer (P.E.) endorsed design calculations and/or drawings shall be required and submitted for special fabricated pipe fittings, pipework supports and thrust restraints.
- (h) In general, all pipes and fittings up to and including 100 mm in diameter shall be manufactured either in steel or in uPVC unless otherwise shown on the 3D model, drawings or required as part of a package plant.
- (i) Diameters of pipes as specified in drawings shall be as follows:
 - (i) Standardised Products: Nominal size;
 - (ii) Fabricated Steel Piping: Inside diameter;
 - (iii) Cement-Lined Steel Pipe: Lining inside diameter; and
 - (iv) FRP Piping: Metric dimensions for inside diameter (ID) series pipe.
- (j) Where steel and cast flanges are mated together, the steel flange shall be machined over its full face after it has been welded to its respective pipe. Wherever possible standard fittings shall be used in preference to fabricated or special fittings.
- (k) Facilities shall be provided for draining the pipe system and releasing air. The drained fluid shall be piped to an approved discharge point.

- (l) Wherever possible, heavy equipment such as valves, flowmeters, strainers and other devices mounted in the pipework shall be supported independently of the pipes to which they connect.
- (m) Unless otherwise specified, flanges shall conform to BS EN 1092 and shall be drilled in accordance with the appropriate pressure designation but generally not less than PN 10 or as specified.
- (n) Higher pressure rated flanges as required shall be used to mate with equipment when equipment flange is of higher pressure rating.
- (o) Where a pipe passes through a reinforced concrete wall, water retaining wall or is subject to thrust, it shall incorporate a puddle flange, which shall conform to the dimensions stated in BS EN 1092 but shall remain undrilled.
- (p) Vertical pipework may be supported on duck foot bends mounted on the civil structure. When used, duck foot bends shall be bolted to the civil structure.
- (q) Metallic flanges shall be raised-face.
- (r) Non-metallic flanges shall be flat-faced with hot-dipped galvanized iron backing rings unless otherwise specified.

G4.2.2 Reference Standards

- (a) Unless otherwise specified and approved by the S.O., pipelines and pipework shall comply with the relevant Singapore Standards and the Reference Standards listed below where applicable:

Standard	Subject
BS 437	Cast iron spigot and socket drain pipes and fittings.
BS EN 124	Manhole covers, road gully gratings and frames of cast iron and cast steel.
BS EN 10311	Steel pipes, joints and specials for water and sewage.
BS 1154	Natural rubber compounds
BS EN 1982	Copper alloy ingots and copper alloy and high conductivity copper castings.
BS EN 1561	Flake graphite cast iron.
BS EN 681 Parts 1 and 2	Elastomeric seals for joints in pipework and pipelines.
BS 2633	Class I arc welding of ferritic steel pipework
BS 2971	Class II arc welding of carbon steel pipework
BS EN ISO 3506 – 1,2,3	Mechanical properties of corrosion resistant steel fasteners.
BS EN 12201	Polythene pipe (Type 50) for cold water services.
BS 3416	Bitumen-based coatings for cold application.
BS EN 1452 PARTS 1 TO 5	uPVC pipes for cold potable water.

Standard	Subject
BS 3506	uPVC pipe for industrial uses
BS EN ISO 15493	Plastics piping systems for industrial applications. ABS, PVC-U and PVC-C. Specifications for components and the system
BS EN 10220	Seamless and welded steel tubes. Dimensions and masses per unit length
BS EN 13598-1	Plastic piping systems for non-pressure underground drainage and sewerage. UPVC, PP and PE. Specifications for ancillary fittings including shallow inspection chambers
BS EN 10216-5	Seamless steel tubes for pressure purposes. Technical delivery conditions. Stainless steel tubes
BS EN 10217-7	Welded steel tubes for pressure purposes. Technical delivery conditions. Stainless steel tubes
ASME B16.9	Factory made wrought butt-welding fittings
ASME B36.10M	Welded and seamless wrought steel pipe
ASME B36.19M	Stainless steel pipe
AWWA C205	Standard for Cement-Mortar Protective Lining and Coating for Steel Water Pipe – 4 Inch (100 mm) and Larger-Shop Applied
AWWA C208-01	Dimensions for fabricated steel water pipe fittings
AWWA C220-07	Stainless steel pipe ½ in. (13 mm) and larger
AWWA C950-07	Fiberglass Pressure Pipe
BS EN 10253	Butt-welding pipe fittings. Non alloy and ferritic alloy steels with specific inspection requirements
BS EN 10253-4	Butt-welding pipe fittings – part 4: Wrought austenitic and austenitic-ferritic (duplex) stainless steels with specific inspection requirements
ASTM A790/A790M	Standard specification for seamless and welded ferritic/austenitic stainless steel pipe
ASTM A999/999M	Standard specification for general requirements for alloy and stainless steel pipe
ASTM A403/A403M	Standard specification for wrought austenitic stainless steel piping fittings
ASTM A995/A995M	Standard specification for castings, austenitic-ferritic (duplex) stainless steel, for pressure-containing parts
ASTM C76/AASHTO M170	Reinforced Concrete Culvert, Storm Drain and Sewer Pipe

Standard	Subject
ASTM C361	Reinforced Concrete Low Head Pressure Pipe
ASTM C655	Reinforced Concrete D-Load Culvert, Storm Drain and Sewer Pipe
ASTM C443/AASHTO M198	Joints for Circular Concrete Sewer and Culvert Pipe, Using Rubber Gaskets
ASTM C497	Testing Concrete Pipe or Tile
ASTM C822	Definitions of Concrete Pipe and Related Products
ASTM C924	Concrete Pipe Sewer Lines by Low-Pressure Air Test methods
ASTM C969	Infiltration and Exfiltration Acceptance Testing of Installed Precast Concrete Pipe Sewer Lines
ASTM D2996-01	Standard Specification for Filament-Wound "Fiberglass" (Glass-Fiber-Reinforced Thermosetting-Resin) Pipe
ASTM D2846/D2846M	Standard Specification for Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Hot- and Cold-Water Distribution Systems
ASTM F2618-15	Standard Specification for Chlorinated Poly (Vinyl Chloride) (CPVC) Pipe and Fittings for Chemical Waste Drainage Systems
ASTM F437-15	Standard Specification for Threaded Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 80
ASTM F441/F441M	Standard Specification for Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedules 40 and 80
ASTM F493-14	Standard Specification for Solvent Cements for Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe and Fittings
ASTM F439-13	Standard Specification for Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 80
BS EN 10216 Parts 1 to 3	Carbon steel pipes and tubes
BS EN 10300	Bitumen based hot applied protective coatings
BS 4190	ISO metric black hexagon bolts, screws and nuts
BS 4320	Metal washers for general engineering purposes, metric series
BS EN 1452 – 1 to 5	Joints and Fittings for uPVC pressure pipes
BS EN 10224	Non-alloy steel tubes and fittings for the conveyance of water or other aqueous liquids

Standard	Subject
BS EN 10029, BS 7668	Weldable structural steels
BS EN 1092 Parts 1,2 and 3	Circular flanges for pipes, valves and fittings: Steel, cast iron and copper alloy
BS 4515 – 1&2	Welding of steel pipelines on land and offshore
BS EN 545	Ductile iron pipes and fittings - for water application
BS EN 598	Ductile iron pipes and fittings – for sewage application
BS EN 14628	Ductile iron pipes, fittings and accessories – External polyethylene coating for pipes – Requirements and methods
BS EN 14901	Ductile iron pipes, fittings and accessories – Epoxy coating (heavy duty) of ductile iron fittings and accessories – Requirements and methods
BS EN 15189	Ductile iron pipes, fittings and accessories – External polyurethane coating for pipes – Requirements and methods
BS EN 15655	Ductile iron pipes, fittings and accessories – Internal polyurethane coating for pipes – Requirements and methods
BS ISO 10804	Restrained joint systems for ductile pipelines – Design rules and type testing
BS EN 1514	Dimensions of gaskets for PN-designated flanges. Non-metallic flat gaskets with or without inserts
BS EN 1171	Cast iron gate valves
BS EN 12334	Cast iron check valves for general purposes
BS 6956	Jointing materials and compounds
BS EN 14364, BS EN 1796	Glass reinforced plastics (GRP) pipes and fittings for use for water supply or sewerage
ASTM D2310	Standard classification for machine-made “fibre glass” (glass-fibre-reinforced thermosetting-resin) pipe
BS PD 5500:2012+A3	Unfired fusion welded pressure vessels
BS 2782 - 11	General requirements for dimensions and pressure ratings for pipe of thermoplastics materials
BS 5911	Precast concrete pipes, fittings and ancillary products
BS 6076	Tubular polyethylene film for use as protective sleeving for buried iron pipes and fittings

Standard	Subject
BS EN 12201 Parts 1 to 5	Blue polyethylene pipes up to nominal size 63 for below ground use for potable water
BS 6956	Jointing of Materials
BS 7079: Part 0	Preparation of steel substrates
BS 8010: Parts 1 & 2	Code of Practice for pipelines: Pipelines on land
API 5L	Specification for Line pipes
API 5LS	Spiral-welded line pipes
API STD 1104	Standard for welding pipelines and related facilities
ASTM A182	Forged, alloy and stainless steel flanges
ASTM A240	Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet and Strip for General Applications
ASTM A380	Standard Practice for Cleaning, Descaling and Passivation of Stainless steel
ASTM A479	Bar Steel
ASTM A480M	Standard Specification for General Requirements for Flat-Rolled Stainless and Heat-Resistant Steel Plate, Sheet and Strip
ASTM A479	Seamless and welded tube and pipe
EN 10088-3	Forged Billets
EN 10216-5	Seamless tube and pipe

(* WRC (UK) Water Treatment Specifications, previously called Information and Guidance Notes)

- (b) This list is not intended to be exhaustive. These standards for equipment or materials provided as part of the works may be applicable to the approval of the S.O..

G4.2.3 Submittals

- (a) Submissions which the Contractor is required to make in relation to pipelines include, where relevant, the following as a minimum:
- (i) Shop Drawings and Data
- 1) Shop Fabricated Piping:
 - Detailed pipe fabrication or spool drawings, showing special fittings and bends, dimensions, coatings, and other pertinent information; and
 - Layout drawings showing location of each pipe section, including the length and diameter; number or otherwise designated laying sequence on each piece.

- 2) Pipe Wall Thickness: Identify wall thickness and rational method or standard applied to determine wall thickness for each size of each different service including exposed, submerged, buried, and concrete-encased installations for Contractor-designed piping;
 - 3) Hydraulic Thrust Restraint for Restrained Joints: Details including materials, sizes, assembly ratings, and pipe attachment methods;
 - 4) Dissimilar Buried Pipe Joints: Joint types and assembly drawings;
 - 5) Manufacturer's calculations, catalogues and data sheets including pressure/temperature ratings for pipes, fittings and joints;
 - 6) For steel pipes, drawings with descriptions outlining the method of forming pipes and specials in standard lengths from steel sheet or strip;
 - 7) Contractor's calculations for pipe and fitting design;
 - 8) Method of boxing out for and building pipes into structure walls;
 - 9) Methods of control of line and level of pipeline during installation;
 - 10) Methods for pipe handling, installation and backfill; and
 - 11) Details of Contractor's pipe supports.
- (ii) Quality Control Submittals
- 1) Chemical Composition of Materials and test reports;
 - 2) Laboratory Testing Equipment or Instruments: Certified calibrations, manufacturer's product data, and test procedures;
 - 3) Weld Inspection and Testing Agency: Certification and qualifications;
 - 4) Welding Inspector: Certification and qualifications;
 - 5) Welders:
 - List of qualified welders and welding operators; and
 - Current test records for qualified welder(s) or welding operator(s) and weld type(s) for factory and field welding. WPQ.
 - 6) Certified welding inspector (CWI) credentials;
 - 7) Weld Procedures: Records shall be maintained in accordance with ASME Boiler and Pressure Vessel Code, Section IX for weld type(s) and base metal(s):
 - Shop and field welding procedure specifications (WPS); and
 - Procedure qualification records (PQR).
 - 8) Testing agency personnel credentials;
 - 9) Non-destructive inspection and testing procedures prepared in accordance with ASME BPVC Section V;
 - 10) Manufacturer's Certification of Compliance:
 - Pipe and fittings;
 - Welding electrodes and filler materials; and
 - Factory applied resins and coatings.

- 11) Certified weld inspection and test reports;
 - 12) Test logs;
 - 13) Full details of internal and external protection and lining systems including method of application, materials, curing, etc.;
 - 14) Pipe coating applicator certification; and
 - 15) Evidence of compliance with requirements for materials, substances and products in contact with water which is to be supplied for drinking.
- (b) Records to be Kept
- (i) The Contractor shall keep detailed and up-to-date records in a form to be approved by the S.O. of all pipes, showing the quantities of each type, size and class which have been:
 - 1) Ordered during the course of the Works;
 - 2) Delivered during the course of the Works;
 - 3) Declared on delivery to be faulty, damaged or deficient;
 - 4) Used in the Works; and
 - 5) Broken, damaged or lost during the course of the Works.
 - (ii) Each week the Contractor shall provide the S.O. with an up-to-date copy of the above records. In the case of steel pipes, the Contractor shall also record the Bill of Quantities item number.
 - (iii) The Contractor shall capture GPS coordinates of the pipe alignment to be updated in the as-built drawings.
- (c) Test Certificates
- (i) The Contractor shall forward to the S.O. before despatch certificates from the manufacturers to show that the pipes have been tested and comply with the appropriate Standard(s) and material compositions.
 - (ii) The Contractor shall forward to the S.O. copies of the certificates of all weld tests carried out on steel pipework.
- (d) Method and Programme of Pipeline Connections, Testing and Cleaning
- (i) The Contractor shall submit for the approval of the S.O.'s details of his proposed methods and programme for pipeline connections including connections to existing pipelines, testing, swabbing, disinfection and cleaning (including details of test and cleaning equipment).

G4.2.4 Material, Workmanship and Welding Standards

- (a) All pipes, flanges, fittings, specials, etc. shall be supplied with manufacturers' guarantees to ensure the suitability of the supplied material for the intended use.
- (b) Material inspection before rolling shall be carried out to check against Mill Certificates and the inspection records shall be submitted. This shall include determination of the country of origin, manufacturer, physical inspection, checking of thickness and visual inspection of edges for lamination.

- (c) As a control on weld quality, the Contractor shall carry out and submit radiographs for 3% of all butt welds for the S.O.'s clearance. The S.O. will advise the Contractor where the radiographs are required to be taken. Radiographs shall be taken daily whenever welding of pipes and specials is in progress. The clearance and acceptance of these radiographs shall be in accordance with API Standard 1104 unless otherwise directed by the S.O..
- (d) Weld defects shall include cracks, leaks, laminations deformity, lack of complete penetration, lack of complete fusion, dents exceeding one eighth of the specified wall thickness and undercutting or reduction in pipe wall thickness adjacent to a weld exceeding 0.5 mm in depth.
- (e) Slag inclusions and gas pockets or voids considered to be minor imperfections may be accepted if the maximum size and distributions does not exceed the limits shown in Section 9 of API Std 5L or Section 7 of API Std 5LS.
- (f) Where radiographic examination reveals defects in the welds the S.O. will either reject the length of pipe containing such defects or will permit the Contractor to carry out repairs and to submit radiographs of such repairs for clearance. Lengths of pipe containing defects in welds after repair shall be liable to be rejected.
- (g) If defects in welds are found in a length of pipe or fitting, the welds of the pipes and fittings immediately before and after the defective pipe or fitting in the production line shall be radiographed until the S.O. is satisfied that all the welds are considered satisfactory. The cost of such radiographs and radiographs of repaired welds shall be borne by the Contractor and shall be included in the Contract rates and shall not form part of the 3% of all welds specified in a previous paragraph of this Clause.
- (h) Hydraulic Testing of Pipes at Place of Manufacture
 - (i) All pipes shall be hydraulically tested at the place of manufacture in accordance with the relevant sections of API Standard 5L or Section 5 of API Standard 5LS as appropriate (as per the Board's standard). Full test pressures of at least 1.5 times the design pressure shall be maintained for at least 3 minutes during testing at the factory and automatic pressure recorders must be installed on hydraulic testing machines. (The design pressure rating for all the delivery pipework shall be not less than 1.5 times the pump shut valve head, or 1.5 times the maximum pipeline operating pressure, whichever is greater.)
 - (ii) Any pipe which fails to withstand the hydraulic test pressure or which exhibits signs of porosity shall be deemed not to comply with this Specification and shall be rejected unless otherwise decided by the S.O..
 - (iii) Unless otherwise specified, physical testing of the pipes shall also be in accordance with the relevant sections of API Standard 5L or Section 7 of API Standard 5LS as appropriate (as per the Board's standard).
 - (iv) All the pipes supplied shall be tested for the intended operating pressure/vacuum conditions at the manufacturers place and a certificate stating the suitability shall be issued.

- (i) Protection of Steel Pipes in Transit
 - (i) All steel pipes shall be protected prior to despatch from the manufacturer's works. All flanges shall have wooden discs bolted on. All other ends of pipes and fittings shall be protected against impact damage and entry of foreign matter. The protection shall take into account the end use intended for the pipes and whether or not the final protection has been completed.
 - (ii) Pipes and fittings shall be wrapped or cushioned so that no load is taken directly on the external coating.
- (j) Delivery
 - (i) The Contractor shall submit to the S.O. advance notices of all consignments of steel pipes. Every consignment shall be accompanied by a detailed delivery note with the item number of each pipe or fitting.
 - (ii) The Contractor shall take delivery of and off-load pipes, valves, and precast manhole sections which he is required to provide, at times and at locations or storage-areas on or about the Site to be arranged by him with his manufacturers/suppliers, subject to the approval of the S.O.. Any such location or storage area which does not form part of the Site shall be arranged for by the Contractor at his own risk and expense.
 - (iii) After pipes, valves and precast manhole sections have been delivered to and off-loaded at delivery locations as aforesaid, the Contractor shall make all arrangements for subsequent transport and handling on or about the Site to the point of installation, including where necessary, any movement into and out of temporary storage.
 - (iv) The Contractor shall inspect all pipes, valves and precast manhole sections upon delivery from the manufacturer/supplier and shall note all damaged, broken or otherwise unsatisfactory items and shall take immediate steps to rectify or replace such broken or damaged or unsatisfactory items. Items with damaged fusion bonded epoxy coatings shall be treated in accordance with the requirements of the general specification section on Painting and Protective Coatings.
- (k) Storage
 - (i) The Contractor shall take into temporary protective storage all pipes, valves and precast manhole sections not required for immediate installation in the Works or, in the case of pipes and precast manhole sections, for stringing out along the pipeline route.
 - (ii) The Contractor shall prepare pipe storage yard areas at location(s) as directed by the S.O. on Site at which the steel pipes and fittings shall be delivered. The areas shall be cleared of vegetation and levelled off and the Contractor shall construct and maintain adequate temporary access to the storage yard(s).
 - (iii) The Contractor shall provide timber pieces to keep all stored pipes clear of the ground at no extra cost to the Board. The Contractor shall maintain the storage yards in a condition satisfactory to the S.O. until completion of the Works.

- (iv) Pipes in storage shall be laid on wedged timber bearers so as to be at least 75 mm clear of the ground; steel pipes shall be stacked in one layer only, other pipes may be stacked up to three pipes high if suitable protective packing is placed between the layers. To prevent damage to the sheathing of pipes which are sheathed with bitumen the bearers shall be positioned at the unsheathed ends of the pipes.
- (v) The period between taking delivery of a pipe and the completion of its installation shall be kept to a minimum and in the case of steel pipes this period shall not exceed six months. The Contractor shall take active measures to ensure that no congestion at site will result from accumulation of stored pipes. Any period during which the pipes are strung out along the pipeline route or placed alongside the Works awaiting installation shall also be kept to a minimum. Jointing parts and materials shall in any case be stored under cover as for valves.
- (vi) Subject to the foregoing and to any restrictions of the duration of temporary occupation of parts of the Site, pipes may be strung out along the pipeline route prior to installation providing that any necessary temporary fencing has first been erected.
- (l) Examination of Steel Pipes prior to Installation
 - (i) All pipes and specials shall be thoroughly brushed through, cleaned and carefully examined for cracks and flaws both internally and externally immediately prior to laying. Materials found to be damaged in any way shall be shown to the S.O. who will give instructions regarding remedial work or otherwise.
 - (ii) Where ends to be joined by welding are coated with ensis oil at the place of manufacture, the ends shall be thoroughly cleaned with the ensis oil removed prior to welding.
 - (iii) The ends of pipes and specials shall be wire brushed if necessary and thoroughly cleaned to remove all traces of rust and dirt.
 - (iv) Should any pipe or specials be found to be damaged, it shall be excluded from the work until repairs have been carried out to the satisfaction of the S.O.. In general, no pipe shall be laid if an end has suffered distortion from circularity and a mechanical coupling or socket cannot be fitted to it or, if damage to the internal and/or external protection has not been satisfactorily repaired.
- (m) Repair of Pipe Linings and Sheathing
 - (i) Only specially competent and skilled worker shall be employed to carry out repair work to damaged linings and sheathing.
 - (ii) Small cracks and flaws in the internal mortar lining of steel pipes and specials may, with the prior approval of the S.O. and if they are accessible, be repaired on Site. Such cracks may be repaired by cutting out a 9 mm x 9 mm dovetailed key and making good with a semi-dry mortar (proportion of 1 part cement to 2 parts coarse sand) well tamped into the groove, and cured by keeping it damp for 7 days. Alternatively, small cracks may be caulked with an epoxy resin filler.

- (iii) The external sheathing of steel pipes and specials shall be made good wherever the steel has been exposed or the thickness of the coating seriously reduced by damage, or where the coating does not adhere tightly to the steel. Where the steel has been exposed, all rust and dirt shall be removed carefully by wire brushing or otherwise, and when perfectly clean and dry, a coat of approved special quick drying primer applied before re-fettling. Small stones which may have become embedded in the sheathing shall be picked out and the coating re-fettled by heating locally with a blow lamp and smoothing over with a fettling knife. Additional external jointing material shall be added wherever necessary.
- (iv) All defects to linings and sheathings shall be made good to the entire satisfaction of the S.O.. Where, as confirmed by the pipe manufacturer and in S.O.'s opinion, the damage is such that satisfactory repairs on Site will not be practicable, e.g. when damage or cracking is extensive and/or inaccessible, the pipes or specials in question shall be either returned to the factory for repairing properly or replaced completely. The Contractor shall be responsible in either case for all costs involved in this respect and for any delays to construction work thereby caused. The compound used for repairing sheathings shall be compatible with the sheathing material and shall be approved by the S.O..

G4.2.5 Qualifications

- (a) Accredited Laboratory
 - (i) Ten years' experience in field of welding and welded pipe and fittings' testing required for this Works;
 - (ii) Calibrated instruments and equipment, and documented standard procedures for performing specified testing;
 - (iii) Certified in accordance with ASNT SNT TC 1A for testing procedures required for this Works;
 - (iv) Testing Personnel: Personnel performing tests shall be NDT Level II Certified; and
 - (v) Inspection Services: Qualified welding inspector.
- (b) Welding Inspector: AWS certified, AWS QC 1 qualified, with prior inspection experience of welds specified.
- (c) Welder and Welding Operator Qualifications
 - (i) Qualified by accepted inspection and testing agency before starting Work in accordance with Section IX, Article III of the ASME Boiler and Pressure Vessel Code;
 - (ii) Qualified to perform groove welds in Positions 2G and 5G or as required for each welding process and pipe material specified; and
 - (iii) Qualification tests may be waived by S.O. based on evidence of prior qualification.
- (d) Retesting: Upon S.O.'s written request, retest qualified welder(s).
- (e) Grooved Piping Installer Qualifications: Five years' experience in field installation of grooved piping systems.

G4.2.6 Quality Control

- (a) Provide services of accredited testing laboratory for welding operations.
- (b) All pipes and fittings manufacturers must be ISO 9001 registered.

G4.2.7 Air Pipework

- (a) Air pipework system shall be well designed and maintained system. The air flow velocity shall be selected to minimise the high pressure loss in the system and prevent any adverse pressure drop and carry over of moisture past the drain legs. The air flow velocity in any pipe or duct shall not be less than 6 m/s. For low air pressure applications of less than 10 Bar, maximum air flow velocity in pipe or duct shall not exceed 30 m/s. For high air pressure applications of 10 Bar or more, this shall not exceed 15 m/s. The design and associated calculations shall be endorsed by P.E. and submitted for S.O.'s approval.
- (b) Metal Pipework shall be earthed to discharge static electricity.
- (c) Drainage points shall be provided and at not more than 30 m intervals. Drainage points shall be formed by equal tees with a down pointing leg fitted preferably where changes of direction occur. The bottom of any falling pipe shall be drained.
- (d) Branch take-offs shall be from the top of the main.
- (e) The pipework system shall be fully hydraulically pressure tested.
- (f) The pipework system shall be designed to avoid resonance and vibration.
- (g) The pipework system shall allow for the thermal expansion and contraction of pipework. Sufficient and suitable bellow type expansion joint with supports, guiding pipe and control rod shall be provided to accommodate the pipe movement. Metallic bellows and guiding pipe, shall be made of stainless steel 316. The details of the pipework arrangement shall be submitted to the S.O. for approval.
- (h) Pipework up to and including 50 mm N.B. shall be in steel to BS EN 10255 with heavy fittings in malleable iron to BS143 screwed to BS EN 10226-1. Threaded joints shall be made with a temperature resistant jointing compound. PTFE tape shall not be used.
- (i) Pipework over 50 mm N.B. shall be in steel to BS EN 10216 grade 360 or 430 with flanged joints drilled to BS EN 1092 PN10 or higher rating where applicable, unless otherwise specified.
- (j) Carbon steel pipework, if specified, shall be hot dip galvanised in accordance with Painting and Protective Coatings General Specification. No other internal coating is required.
- (k) Sufficient bellow type expansion joint with supports, guiding pipe and control rod shall be provided to accommodate the pipe movement.
- (l) Particular care shall be taken at points where condensation is likely to form or accumulate such as following compression or cooling of ambient air or discharge of reactivation air from dryers.
- (m) Appropriate traps and automatic drain valves discharging to approve points shall be installed.

- (n) Unless otherwise specified, thermal insulation and cladding or protection guard where fitted shall be arranged in a manner which provides access to valves and other components for maintenance purposes.
- (o) Pipework connected to blowers shall be via flexible connectors, with the pipework immediately downstream of the flexible joint being suitably supported.
- (p) Test points shall be provided at the outlet and inlet pipework and shall be proposed and submitted to S.O. for approval.
- (q) Wall penetrations and pipework supports shall be designed to limit the transmission of noise/vibration.
- (r) Equipment foundations, supports and restraints, etc. shall be designed and installed with due regard to all static and/or dynamic forces that will be imposed upon them by the installed equipment including, as a minimum, forces arising due to surge, thermal effects and vibration.

G4.2.8 Pipework Systems Conveying Sewage or Sewage Derivatives

- (a) All pipework for sewage or liquid sewage derivatives shall be designed and constructed such that blockage clearance is possible for the entire pipework system.

G4.2.9 Pipework Systems Conveying Liquids

- (a) Pipework systems that convey liquids shall:
 - (i) Include facilities and all equipment needed for priming, venting, flushing and draining;
 - (ii) Wherever possible, be vented by continuous rise. Where continuous rise is not possible, automatic air vents shall be provided at high points in the pipework system and vented as appropriate;
 - (iii) Be protected from the effects of freezing at temperatures down to an ambient air temperature of -10°C (or lower where the site is exposed to wind chill), e.g. by insulation and, where needed, trace heating;
 - (iv) Be insulated to protect against harm to personnel where pipe contents are either hot or very cold; and
 - (v) Be sized such that the mean fluid velocity does not exceed 2.5 m/s and according to the specific velocity criterion in the respective contracts' Particular Specifications. However:
 - 1) For suction pipes the velocity shall not exceed 1.5m/s; and
 - 2) For infrequent occurrences, or short distances, the maximum instantaneous velocity shall not exceed 2.7m/s for gravity flows and 3.7m/s for pressurised pipelines.
- (b) Flushing and drain connections on pipework below 100 mm shall be made using proprietary welded fittings to BS EN ISO 228-1. These fittings shall be sealed immediately with hexagon headed shouldered plugs and seals. Holes made in pipes shall have all burrs removed and be cleaned through to remove loose particles.
- (c) Gravity flow pipelines for any process fluids containing settleable solids shall be laid at a gradient to ensure self cleansing velocities are achieved. Unless specified elsewhere, the Contractor shall comply with the permissible velocities and grades stated in the PUB Code of Practice on Sewage and Sanitary Works.

G4.2.10 Pipework Systems Conveying Gases

- (a) Pipework systems for non-flammable gases shall be provided with facilities and all equipment needed for purging, venting and condensate draining.

G4.2.11 Pipework Materials

This Section is specific to the material requirements for pipework.

Pipework shall be manufactured to the specification given hereunder and shall be provided by the Contractor except as specified otherwise. The particular type, size and class of pipes for each part of the Works shall be as specified or as described on the 3D model, Drawings or in the Bills of Quantities.

- (a) Steel Pipework
 - (i) Pipes shall be specified by nominal diameters in accordance with BS EN 10220.
 - (ii) Carbon steel pipes shall be in accordance with BS EN 10216 parts 1 to 3 or BS EN 10217 parts 1 to 3. Carbon steel fittings shall comply with BS EN 10253-1.
 - (iii) Welding shall be carried out in accordance with BS 2633 Class 1 arc welding. All welders shall be tested by an independent inspection authority and shall satisfy the requirements of BS EN 287-1. Current and valid welder test certificates shall be provided by the Contractor for approval by the S.O. prior to commencement of site welding. The S.O. may reject any proposed welder on the basis of incompetence at any time during the implementation of the Works.
 - (iv) Steel pipework fittings shall be fabricated in accordance with BS EN 10311. Where dimensions for the required fittings are not given in BS EN 10311, AWWA Standard C208 for "Dimensions for Fabricated Steel Water Pipe Fittings" shall be used.
 - (v) All fabrication and corrosion protection shall be carried out at the manufacturer's works to the furthest extent possible to limit site fabrication but maximise site installation works. Linings and coatings shall comply with the general specification section on Painting and Protective Coatings.
- (b) Internally lined Steel pipes
 - (i) Steel pipes shall be manufactured, except where stated otherwise hereunder, in accordance with API Specification 5L or API Specification 5LS. Material for pipes shall be carbon steel plate to Grade 43A of BS EN 10029 or better with a minimum lower yield stress of 275 N/mm² for plate thickness up to 16 mm and 265 N/mm² for plate thickness greater than 16 mm and up to 40 mm, and shall be supplied in plate or strip form as appropriate for the method of pipe manufacture. The pipes shall be formed by the automatic submerged arc process, with either a longitudinal seam or a spiral seam at the option of the Contractor. With the method of manufacture decided upon, the relevant standard shall then apply i.e. API Specification 5L for longitudinal seam pipe or API Specification 5LS for spiral seam pipe. The API standards shall be referred to hereunder as Std 5L/S and shall be deemed to apply to all pipe sizes, including those whose outside diameter (OD) falls outside the upper limit covered by the standards.

- (ii) The Contractor shall provide a ladle analysis of the steel used for the pipes. The check analyses shall be based on BS 7668, BS EN10029 , BS EN 10311 and BS EN 10025. In case of dispute, the methods of chemical analysis shall be in accordance with BS Handbook No. 19.
- (iii) Unless otherwise specified, thickness of the steel plates and the finished outside diameters of the pipes shall be as shown in the following table:

Nominal Size (mm)	Thickness of steel plate (mm)	Thickness of concrete Lining (mm)	Outside diameter of pipe (mm)
2200	16	25	2235
2000	16	25	2032
1900	16	25	1930
1800	12.7	25	1829
1700	12.7	25	1727
1600	12.7	25	1626
1500	12.7	25	1524
1400	12.7	25	1422
1300	12.7	25	1321
1200	12.7	19	1219
1100	12.7	19	1118
1000	12.7	19	1016
900	9.5	19	914
800	9.5	19	813
750	9.5	19	762
700	9.5	13	711
600	9.5	13	610
500	9.5	13	508
450	6.3	13	457
400	6.3	13	406.4
300	6.3	10	323.8
250	6.3	10	273
200	6.3	10	219.1
150	6.3	6	168.3
100	6.3	6	114.3
*The above thickness of steel plates shall not be subject to any negative tolerance for acceptance purpose.			

- (iv) The thickness of pipes used for branch pipework off the main pipelines or for use in forming fittings shall be the greatest of the following:
- 1) The thickness necessary to provide the same outside diameter to wall thickness ratio specified for straight pipes;
 - 2) The minimum thickness shown in Table 9 of Std 5L or Table 6.1 of Std 5LS for pipes of the appropriate diameter (or the nearest equivalent therein); and

- 3) The thickness calculated and determined to be necessary in the design of fittings as shown on the 3D model or Drawings.
- (v) Standard length straight pipes shall be manufactured with not more than three circumferential joints and with an effective length as shown in the table below:

Finished internal dia (mm)	Standard Length (m)
More than 2100	9
450 to 2050	10
Below 450	6

- (vi) The allowable tolerance on standard length pipes shall be ± 150 mm. In addition, a number of straight pipes of half the standard length pipes and truly circular standard straight pipes shall be supplied. All standard and half-length pipes shall have ends formed as specified below for joints capable of angular rotation. Truly circular standard straight pipes required for cutting purposes shall be truly circular throughout the length.
- (vii) All pipes whether manufactured with a longitudinal or spiral seam shall be subjected to non-destructive testing of seam welds and skelp and welds in accordance with the relevant sections of Std 5L or Section 7 of Std 5LS as appropriate.
- (c) Stainless Steel pipes
- (i) Stainless steel pipe shall conform to the relevant applicable design standard codes listed under section G4.2.2 (a) in this specification and any other relevant latest standards. It shall be compatible with the fluid conveyed, be of the minimum required Pitting Resistance Equivalent Number (PREN), having pipe wall thickness schedule taking account of pressure rating and whether required for butt-welding or threading or other end connections and be of the required grade.
- (ii) Requirements for stainless steel pipes for major applications of design pressures of 10 Bar and 16 Bar are summarized as follows:

Material Classification SS-1

Pipe Classification: SS-1	Pipe Material: Stainless Steel
Service: (Refer to the Particular Specifications for exact applications)	
Design Pressure: 10 Bar	Test Pressure: 15 Bar
Design Temperature: 15 to 240 °C	Test: Hydrostatic
Material Spec.: ASTM A312, Grade TP 316L / A240-Type 316L	
Gasket thickness: To be full-face 3 mm thick. Gasket material options: a.) Nitrile rubber for up to 108 °C applications only. b.) Neoprene rubber and EPDM rubber for up to 120 °C applications only. c.) PTFE and fibre reinforced EPDM for up to 204 °C applications only. d.) Natural rubber for up to 220 °C applications only. e.) Viton rubber for up to 250 °C applications only. Note: Contractor shall submit gasket manufacturer's datasheets with certification of the temperature ratings and suitability of application for the specific fluids in contact with the gaskets. This is especially in view of material variants that have inherent properties modified to increase performance.	
Bolts: Type SS316, Dimensions per BS 4190 and BS EN ISO 3506-2	
Nuts and Washers: Type SS316, Dimensions per BS 4190 and BS EN ISO 3506-2	

Nominal Pipe Size, mm	Schedule (1) / Min Wall Thickness, mm	Ends	Flanges	Fittings	Lining	Coating
15	SCH40S	Plain End	Socket welded A 182, F316L, 150lbs (68Kg) RF	Socket welded A 182, F316L, 3000lbs (136kg)	None	None
20						
25						
40						
50	SCH 10S	Bevelled End	Stainless steel plate, Type 316 To match mating flange where indicated BS EN 1092-1 PN 10 Rated Flange where none is indicated	Butt welded ASTM A403, WP316L. Fittings shall conform to, ANSI/ASME B16.9 with bevelled ends Where dimensions for the required fittings are not given in ANSI/ASME B16.9, AWWA Standard C208 will be used.		
65						
80						
100						
150						
200						
250						
300						
350						
400						
450						
500						
550						
600	6 mm					

Nominal Pipe Size, mm	Schedule (1) / Min Wall Thickness, mm	Ends	Flanges	Fittings	Lining	Coating
700	6 mm		SO Flange Flat Face	Threaded connections should be kept to a minimum, where required shall be BSP		
750	6 mm					
800	6 mm					
900	6 mm					
1000	6 mm					
1100	6 mm					
1200	6 mm					
1300	8 mm					
1400	8 mm					
1500	8 mm					
1600	8 mm					

Note:

1. Pipe Schedule Dimensions as per ASME B36.10.
2. Applications requiring threading shall use wall thickness schedules of greater than SCH. 10S.

Material Classification SS-2

Pipe Classification: SS-2	Pipe Material: Stainless Steel
Service: (Refer to the Particular Specifications for exact applications)	
Design Pressure: 16 Bar	Test Pressure: 24 Bar
Design Temperature: 15 to 240 °C	Test: Hydrostatic
Material Spec.: ASTM A312, Grade TP 316L/ A240-Type 316L	
Gasket thickness: To be full-face 3 mm thick. Gasket material options: a.) Nitrile rubber for up to 108 °C applications only. b.) Neoprene rubber and EPDM rubber for up to 120 °C applications only. c.) PTFE and fibre reinforced EPDM for up to 204 °C applications only. d.) Natural rubber for up to 220 °C applications only. e.) Viton rubber for up to 250 °C applications only. Note: Contractor shall submit gasket manufacturer's datasheets with certification of the temperature ratings and suitability of application for the specific fluids in contact with the gaskets. This is especially in view of material variants that have inherent properties modified to increase performance.	
Bolts: Type SS316, Dimensions per BS 4190 and BS EN ISO 3506-2	
Nuts and Washers: Type SS316, Dimensions per BS 4190 and BS EN ISO 3506-2	

Nominal Pipe Size, mm	Schedule (1) / Min Wall Thickness, mm	Ends	Flanges	Fittings	Lining	Coating
15	SCH40S	Plain End	Socket welded A 182, F316L, 150lbs RF	Socket welded A 182, F316L, 3000lbs	None	None
20						
25						
40						
50						
65	SCH 10S	Bevelled End	Stainless steel plate, Type 316 To match mating flange where indicated BS EN 1092-1 PN 16 Rated Flange where none is indicated SO Flange Flat Face	Butt welded ASTM A403, WP316L Fittings shall conform to, ANSI/ASME B16.9 with bevelled ends Where dimensions for the required fittings are not given in ANSI/ASME B16.9, AWWA Standard C208 will be used. Threaded connections should be kept to a minimum, where required shall be BSP		
80						
100						
150						
200						
250						
300						
350						
400						
450						
500						
550						
600						
700						
750						
800						
900						
1000						
1100						
1200						
1300						
1400						
1500						
1600						

Note:

1. Pipe Schedule Dimensions as per ASME B36.10
2. Applications requiring threading shall use wall thickness schedules of greater than SCH. 10S

(d) Ductile iron pipes and fittings

- (i) Pipework and fittings shall comply with BS EN 545 for water service and BS EN 598 for sewage application.
- (ii) Flanges shall be either cast or welded, or screwed on dependent upon the class designation of pipe required. Flanges shall be rated PN-10 or PN-16 as appropriate unless otherwise stated.
- (iii) Except where indicated otherwise, all socket and spigot pipes and fittings and flanged pipes and fittings shall be to the standard class designation (e.g. 20, 25, 30, 40, 50, etc.).

- (iv) Pipework shall be hydrostatically tested to BS EN 545 and BS EN 598 and a manufacturer's certificate submitted to the S.O..
- (v) Where glass lining has been specified, the following shall comply as minimum:
 - 1) The lining shall completely fused above 760 degrees C, 0.1524 mm to 0.254 mm thick;
 - 2) The defects which expose base metal shall not be greater than 0.01 % of total lined surface;
 - 3) Hardness shall be greater than 5 on the Mohs scale;
 - 4) The lining bonded shall sufficiently withstand a metal strain without damage to the glass lining; and
 - 5) The finished lined pipe shall not deviate more than 1 mm per metre of length from a centerline perpendicular to the flange face or square end of the pipe.
- (e) Unplasticised Polyvinyl Chloride (uPVC) Pipes
 - (i) uPVC pipework for pressurised services shall be of the 'hi-impact' type to BS EN 1452 PARTS 1 TO 5 class E with solvent welded joints and fittings complying with the relevant parts of the specification;
 - (ii) uPVC pipes, joints and fittings for drains shall comply with the relevant provisions of BS 4660 or BS EN 1401-1;
 - (iii) uPVC pipes used as cable ducts shall comply with BS 3506 Class E;
 - (iv) Solvent cements for jointing uPVC pipes shall comply with BS EN 1452. For pipes and fittings complying with BS 4660, solvent cement may alternatively comply with BS EN 14680;
 - (v) All uPVC pipes for process pipework and sewerage purposes shall comply with ASTM D1785, BS 3506 and BS 2782–11 as appropriate, and shall have a minimum sustained pressure rating of 12 bar;
 - (vi) Screwed connections to metal components shall be made with proprietary iron to uPVC adaptor fittings;
 - (vii) Where flanged joints are required, full face uPVC flanges and galvanised mild steel backing rings shall be used;
 - (viii) Where uPVC pipework systems connect to steel mains, the off-take points shall be isolated by means of gate valves to BS EN 1171 or BS 5163. Other valves in uPVC pipework systems shall be as specified under the general specification section on Valves, Penstocks and Stoplogs or as specifically approved by the S.O. with regard to function, location and operating conditions; and
 - (ix) Pipework shall be supported in accordance with manufacturer's recommendations and adequate provisions shall be made for thermal expansion and contraction.
 - (x) uPVC pipework systems shall be protected from exposure to ultra-violet radiation (sunlight). If stored outdoors, the pipework shall be covered with opaque sheeting. If installed outdoors, the pipework shall be painted with water-based UV resistant paint or insulated.

- (f) Chlorinated Polyvinyl Chloride (cPVC) Pipes
- (i) cPVC piping systems design shall generally conform to BS EN ISO 15493 or ASTM F441;
 - (ii) cPVC pipe dimensions shall comply with the requirements of ASTM F441 whilst cPVC pipe socket dimensions shall comply with the requirements of ASTM F439;
 - (iii) The pressure rating of cPVC pipework shall be derated at elevated temperatures according to ASTM F441, or an equivalent standard.
 - (iv) Components for external above-ground installations shall be adequately protected against UV radiation; and
 - (v) A particular category of application of cPVC pipes and fittings may be for the design pressure of 6 Bar and the details for these are as follows:

Material Classification CPVC1

Pipe Classification Datasheet	
Pipe Classification: CPVC1	Pipe Material: PVC(C)
Service: (Refer to the Particular Specifications for exact applications)	
Design Pressure: 6 Bar	Test Pressure: 9 Bar
Design Temperature: 15 to 40 °C	Test: Hydrostatic
Material Spec.: BS EN ISO 15493	
Gaskets: Full face, 3 mm thick, NBR Rubber	
Bolts: A4-70 per BS 4190 and BS EN ISO 3506-2	

Nominal Pipe Size, mm	Schedule / Min Wall Thickness, mm	Ends	Flanges	Fittings	Lining	Coating
15	Thickness of pipes shall be at least in accordance with Schedule 80 as specified in ASTM F441	Plain End	FF, Diameter and drilling shall conform to BS EN 1092 or ANSI/ASME B 16.5, Class 150lbs (ensure compatibility when connecting with flange of equipment or metallic pipework)	ASTM F493, Cell Classification 23447, bearing NSF seal.	None	None
20						
25						
40						
50						
65						
80						
100						
150						
200						
250						
300						
350						
400						

- (g) Where any other application category of CPVC pipes and fittings is required, these are identified with elaboration on their requirements in the respective unit processes' particular specifications.
- (h) High Density Polyethylene (HDPE) Pipes
 - (i) High density polyethylene pipes and fittings for pressure pipelines shall be in accordance with BS EN 12201 Parts 1 to 5.. The manufacturer shall be responsible for all testing and quality assurance procedures and be covered by a third party certificate scheme all in accordance with these standards.
 - (ii) Pipes exceeding 90 mm nominal diameter shall not be supplied in coils.
 - (iii) The Contractor shall ensure that all HDPE pipes and fittings for thermal fusion jointing to form an individual pipeline or pipeline systems shall be supplied by one manufacturer, unless the S.O. approves otherwise. Electrofusion couplings are exempted from this requirement but must be of a type approved by S.O. and shall incorporate pins which pop up to confirm that optimum melt pressure has been achieved.
 - (iv) Mechanical joints shall be complete with liners.
- (i) Fibreglass Reinforced Plastic (FRP) Pipe
 - (i) FRP pipes shall conform to the relevant applicable design standard codes as listed under section G4.2.2 (a) with requirements summarized as follows:

FIBREGLASS REINFORCED PLASTIC (FRP) PIPE		
Item	Size	Description
Pipe	All Sizes	Fibreglass reinforced plastic, helically wound, conforming to ASTM D2310 Type 1, vinyl ester resin, Type C glass monofilament surfacing mat and/or Dynel organic fibre synthetic surfacing veil or approved equal. Constructed in conformance with ASTM D2996 UV Protection: Add to the wax coat
Fabrication	All Sizes	Interior surfacing veil shall be 250 to 500 microns thick. Mat layers shall be resin-rich, fully wetted, 1.3 mm minimum thickness
		One layer veil and two mat layers bell-and spigot, butt and wrap and flanged
Ratings		Fibreglass reinforced pipe, fittings, and flanges shall be rated accordingly to at least 1.5 times maximum operating pressure of the pipework
Minimum continuous strand glass filament (coated with resin) wall thickness for pipe	50 to 150 mm 200 mm 250 mm to 300 mm 350 mm 400 mm to 450 mm 500 mm 600 mm	4.75 mm 6.35 mm 7.95 mm 9.50 mm 11.10 mm 12.70 mm 14.30 mm

FIBREGLASS REINFORCED PLASTIC (FRP) PIPE		
Item	Size	Description
Joints	50 mm to 400 mm Greater than 400 mm	Adhesive bonded/O-Ring tapered bell and spigot. Butt and wrap Flanged at valves and equipment and where required for ease of dismantling to suit situation
Fittings	All sizes	Fittings: ASTM C582 and D2996. Surface veil and mat layers same as specified above under Ratings Bends shall be long radius Structural body of fittings shall be filament wound or built-up with alternate layers of woven roving and chopped stand mat. Filament wound fittings shall be same thickness as specified above under Fabrication
Flanges	All Sizes	Fibreglass reinforced plastic, flat-faced and drilled BS EN 1092 of the appropriate rating specified under Ratings or to match the equipment flange
Bolting		Bolting, nut and washer material shall be A4-70 per BS EN ISO 3506
Gaskets		Flat-Face Flanges: Ethylene propylene rubber (EPR), 3 mm thick, full-faced
Colour		Add pigment to the final layer only of exterior surfacing resin to lightly tint the surface, but not obliterate laminate quality. Colours are as follows: Green

(j) Super Duplex Stainless Steel Pipe

(i) General

- 1) The design, manufacture and testing of the Super Duplex stainless steel pipes, fittings, other piping elements and equipment materials shall conform to the latest versions of international codes, local statutory regulations and codes.
- 2) In addition, compliance with all statutory safety requirements and the OHS Act and Regulations are required.
- 3) Quality controls for the review, verification and management of the design and manufacture of the pipes shall be planned, implemented and documented by appropriately skilled and qualified people. ISO 9001 Quality Assurance Standard certification shall be in place. This shall ensure that the design requirements are met and that a high quality standard is maintained.
- 4) All components and materials shall be new and of the best quality to suit the site condition ensuring reliable and safe operation and facilitating inspection, cleaning and maintenance.
- 5) This Specification is applicable for the high pressure rating pipework associated with process equipment as defined in the tender documents.

(ii) Shop Drawings

- 1) It is the Contractor's responsibility to ensure that all the pipe sections, pipe fittings and pipe specials fabrications are fully detailed on the Shop Drawings and that these comply with the Contract documents. Any acceptance done by the S.O. shall not relieve the Contractor of this responsibility.
- 2) Shop drawings shall be prepared in accordance with sound engineering practice having due consideration to flexural and deflection limits, structural adequacy of member connections, concentric intersections of member forces, differential expansion, pressure requirements, installation requirements, etc.
- 3) The Shop drawings shall show the relevant details of each assembly, component and connection, together with information relative to fabrication and erection, including:
 - Identification mark numbers;
 - SAF 2507 Super duplex material identification;
 - Dimensions of items;
 - Pressure rating;
 - Required camber, where applicable;
 - Fabrication methods including, where applicable, hot or cold forming and post weld heat treatment;
 - Location, type and size of welds and bolting;
 - Weld categories and bolting categories;
 - Orientation of members;
 - Location of temporary connections;
 - Procedures necessary for workshop trial assembly and site assembly;
 - Lifting and support points for handling, transport and installation into tank; and
 - Temporary bracing, if required for handling and transport.

(iii) Performance Guarantees

- 1) All pipe elements and equipment shall comply with the documents quoted; and
- 2) The pipe elements and equipment guarantee terms and conditions are set out in the Terms and Conditions of the Contract Documents.

(iv) Materials of Construction

- 1) Unless otherwise nominated, the equipment in this technical specification shall all be fabricated in Super Duplex stainless steel grade SAF2507.

- 2) All Super Duplex shall have a minimum Pitting Resistance Equivalent Number (PREN) above 40 and Crevice Factor (CF) above 38 as calculated as follows:

$$\text{PREN} = \text{Cr (\%)} + 3.3 \times \text{Mo (\%)} + 16 \times \text{N (\%)}$$

$$\text{CF} = \text{Cr (\%)} + 3 \times \text{Mo (\%)} + 15 \times \text{N (\%)}$$

(v) Coatings

- 1) Super Duplex materials do not require any surface treatment.

(vi) Design criteria

- 1) All the fabrications shall be in Super Duplex SAF2507 unless otherwise specified. The Supplier shall detail the design and fabrication methods proposed based on the limited range of section profiles available in SAF2507. The design and fabrication methods shall be approved by the Board or the S.O. prior to the commencement of any work.
- 2) The Contractor may propose the adoption of alternative standards giving comprehensive details and explanation subject to the Board or S.O.'s approval. The use of alternative standards shall be subject to the agreement and approval by the Board or S.O. prior to manufacture.

(vii) Welding and Passivation

- 1) All welding shall comply with the requirements found in subsequent sections of this specification and according to the SAF2507 suppliers' recommendations. All welds shall be continuous fillet welds. The fabrication design is to minimise site welding as much as practically feasible.
- 2) All stainless steel welding shall be post weld pickled/passivated in accordance with ASTM A380.

(viii) Fasteners

- 1) All bolts, set screws, pipe flange stud bolts, Chemset anchors, washers, nuts and shall be in SAF2507 (grade S32750) unless otherwise noted. They shall have metric threads conforming to ISO metric or BS EN standards. Thread profiles shall be formed by rolling and shall be passivated after forming to reduce the propensity of threads to gall.
- 2) All pipe studs shall be in SAF 2507 (grade S32750) and shall have metric threads. Thread profiles shall be formed by rolling and shall be passivated after forming to reduce the propensity of threads to gall. Each stud shall be fitted with 2 hexagonal head nuts.
- 3) All Chemset anchors shall be in SAF 2507 (grade S32750) and shall have metric threads. Thread profiles shall be formed by rolling and shall be passivated after forming to reduce the propensity of threads to gall. Each anchor shall be supplied with a hexagonal head nut and washer.

(ix) Pipe Gaskets

- 1) All pipe flange connections are to be supplied complete with gaskets, unless otherwise specified for specialized applications.

- 2) The gaskets are to be full face, 3 mm thick in EDPM material.
- (x) Installation
- 1) The Supplier shall provide a detailed installation manual that will allow the mechanical installation contractor to install the equipment and pipework concerned.
- (xi) Inspection and Testing Plan
- 1) The Supplier shall prepare a detailed and comprehensive inspection and testing plan (ITP) including checklists for the Contractor's use during the project. The ITP shall cover all activities depicted in the tender documents, 3D model and drawings and be submitted for approval prior to commencing work. The S.O. will review the ITP and may nominate 'Hold Points' for inspection and monitoring. The revised ITP which has incorporated the S.O.'s review comments shall be resubmitted for final approval.
 - 2) The testing programme submitted to the S.O. for acceptance shall set out the sequence, approximate timetable and methods of testing to be used. The S.O. shall be given at least 28 days' advance notice, or as otherwise agreed in the revised ITP, of such testing so that the S.O.'s representative may elect to attend these tests.
 - 3) All test data shall be submitted to the S.O. in accordance with the Manufacturer Data Requirements.
- (xii) Inspection Requirements
- 1) As a minimum, the following inspections shall be conducted on the components:
 - Design compliance (as part of the Manufacturer data review);
 - Material certification and material compliance (e.g. for all plate, bar, seamless and welded tube and pipe, flanges, billets and fasteners);
 - Welding and compliance to weld procedures (and heat treatment procedures where applicable) shall be reviewed;
 - Welding non-destructive examination (10 % of all butt and fillet welds as a minimum if not nominated in any relevant standard);
 - Dimensional and tolerance compliance including all formed sections during fabrication;
 - Specification and Standard compliance during fabrication; and
 - 100 % Positive Material Identification testing.
- (xiii) Inspection and Test Methods
- 1) The qualification of welding procedures and personnel shall be completed prior to the commencement of welding of the fabrications.
 - Welding procedures shall be qualified in accordance with the requirements found in subsequent sections of this specification; and

- Qualification of the Welding Personnel shall be in accordance with approved Welders' Qualification Procedures.
- (xiv) Trial Assemblies
 - 1) Trial assemblies shall be identified subject to approval and made available for inspection and approval by the S.O. or S.O.'s appointed inspector.
- (xv) 'Hold Points'
 - 1) The 'Hold Points' items are to be submitted for review and approval. No work is to proceed until 'Hold Points' have been approved by the S.O..
- (k) Vitrified Clay Pipes
 - (i) Vitrified clay pipes and pipeline fittings shall comply with the relevant provisions of BS 65 or BS EN 295; and
 - (ii) Vitrified clay pipes shall be of 'normal' type with flexible mechanical joints.

G4.2.12 Pipe Cutting

- (a) Where it is necessary to cut pipes to provide closing lengths or for laying pipe specials or for the repair of damaged or chipped pipes, the cutting shall be neatly and accurately performed so as to leave the end of the pipe truly normal to the axis of the pipe.

G4.2.13 Closing Lengths

- (a) General
 - (i) The use of closing lengths shall be avoided by laying the pipes in a continuous length. Pipes which are required to be cut to form closing pieces in any section of the pipeline or to terminate in manholes or other parts of the Works shall not be cut until after all adjacent pipes have been installed and jointed. Pipes for closing lengths shall be cut to allow a 20 mm gap between adjacent pipe ends.
 - (ii) The Contractor shall determine the length of each closing piece and the end of the pipe shaped up and trimmed so as to ensure an accurate joint or termination as the case may be. Any damage to sheathing coating or lining shall be made good.
 - (iii) Pipes terminating in manholes and the like shall except where otherwise specified be cut so that the end of the pipe is flush with the face of the structure into which it is built.
 - (iv) The closing joints in pressure pipelines shall be made with detachable flexible couplings with locating plugs.
 - (v) The Contractor may arrange at his own risk for cuts to be performed by the pipe supplier.

- (b) Steel Pipes
 - (i) Where closing lengths are required the Contractor shall use the special 'sized' pipes for cutting to the exact lengths required. These steel 'sized' pipes manufactured truly circular along their full length shall be marked by two longitudinal bands in paint along each pipe. Closing lengths shall be located as near to access openings as possible so as to enable internal and external welding at joints to be carried out.

G4.2.14 Internally Lined Fittings

- (a) All bends, tees and other fittings shall be prefabricated and tested by the pipe manufacturer in accordance with BS EN 10311 and BS EN 10224.
- (b) Fittings shall be formed from completely lined and hydraulically tested straight pipes as specified unless otherwise approved by the S.O., with suitable rubber spacers inserted during the lining operation. The coating and lining on the straight pipes shall then be cut back from the ends to be welded or cut for a sufficient distance to ensure that no material which is intended to remain part of the coating/lining is damaged or affected by the welding or cutting process.
- (c) Welding shall be of a standard equal to that specified for straight pipes. All butt welds shall be subjected to a 100 % radiographic test. All fillet welds shall be subjected to air tests where appropriate and to 100 % magnetic particle tests.
- (d) Branches of tees shall be made from the same material as the main body.
- (e) The Contractor shall design and reinforce steel fittings to withstand the specified pressure test and to enable self-support in accordance with the methods contained in the American Water Works Association (AWWA) Steel Pipe Design Manual, BS PD 5500 or by other methods approved by the S.O.. When using the AWWA M11 method, the test pressure for fittings shall be one and a quarter times the design pressure and when using the BS PD 5500 method, the test pressure shall be the "standard" test pressure determined in accordance with BS PD 5500. The design pressure shall be taken as at least one and a half times the specified maximum working pressure.
- (f) Prior to fabrication of any fitting, the Contractor shall have obtained the S.O.'s approval of the calculations, drawings and fabrication procedures for that fitting.
- (g) The outside diameter of specials shall conform to the outside diameters of the standard straight pipes. The ends of plain ended specials shall be truly circular and shall conform to the tolerances required for the fitting of mechanical couplings and flange adaptors.
- (h) Ring girders shall be provided to conform with details shown on the Drawings. They shall be supplied complete with all accessories. Metallic parts are to be descaled to second quality standard according to BS 7079 and BS EN ISO 8501-1 and painted with two pack polyamide cured epoxy zinc phosphate primer with 75 microns minimum dry film thickness.

- (i) Retaining Rings for Internal lining
 - (i) All flanged pipe ends, flanged branches and plain ends for use with mechanical couplings or flange adaptors shall have a 6 mm steel retaining ring welded into the bore of the pipe flush with the end of the pipe after which the ring shall be shot blasted to SA 2½ to BS 7079 followed by the coating system (eg. System 22, etc.) as specified in the General Specification – Painting and Protective Coatings. The radial thickness of the rings shall be similar to the thickness of the concrete lining and shall not be less than 6 mm.

G4.2.15 Jointing

- (a) General
 - (i) Pipe joints shall be in accordance with BS EN 10311 and as described below with the amendments stated.
 - 1) Pipes jointed in the trench or above ground shall have sleeve welded (spigot and socket) joints of the hemispherical type in accordance with BS EN 10311 unless otherwise specified; and
 - 2) For changes in pipe grade or direction, the joints may have angular deflections up to 5 degrees provided that the joint design is suitable and will allow the tolerance for welding to be maintained. The minimum overlap of pipe ends at maximum deflection shall be four times the pipe thickness
 - (ii) For spigot and socket type joints of the hemispherical type, the maximum angular deflection shall be 2 degrees.
 - 1) Butt welded joints may be used as an alternative to sleeve welded joints for pipes to be jointed outside the trench subject to the approval of the S.O.;
 - 2) Detachable flexible couplings shall be used where specified or where shown on the 3D model or Drawings;
 - 3) Flange joints shall be used where shown on the 3D model or Drawings; and
 - 4) Welding collars in accordance with BS EN 10311 shall be used for jointing cut pipes or closing lengths in trench.
 - (iii) Spigot and socket joints shall be of the hemispherical sleeve type with a minimum penetration of the formed ends as specified below. The joint design shall provide for the contact surface of the spigot end and the sleeve end to be formed to the same spherical radius which shall not be less than 50 % of the outside of the barrel of the pipe for the hemispherical type.
 - (iv) Pipes to be used in conjunction with welded collars, mechanical couplings or flange adaptors shall be supplied with plain ends.
 - (v) Sleeve welded joints shall have external fillet welds on pipes of 710 mm outside diameter and under but shall have both internally and external fillet welds on pipes with an outside diameter greater than 710 mm.

- (vi) The gap between sleeve and spigot at any weld location for internal and external welds on Type 3 joints (i.e. hemispherical) shall nowhere exceed 1.5 mm whether the joint is deflected or not and the average gap around the circumference of any weld shall not exceed 1 mm.
- (vii) The minimum penetration of the spigot into the socket shall not be less than four times the pipe thickness and this overlap shall be obtained at the maximum deflection of 2 degrees. At zero deflection the minimum overlap shall be the minimum penetration as specified above increased by such allowance as is necessary to ensure contact between the spigot and socket over the specified maximum amount of angular deflection at the joint.
- (viii) Pipe ends of the hemispherical type shall be formed by hydraulic pressing using a full circle die or expanding former capable of forming ends consistently to a constant spherical contact surface throughout the pipe production run.
- (ix) Fillet welds shall be strength welds having a throat thickness of not less than 0.7 times the pipe thickness. Butt welds shall be full penetration welds having a throat thickness of not less than the pipe thickness.
- (x) The internal surface of sleeves and the external surface of the spigots shall be ground smooth along the longitudinal fusion welds for a distance of 150 mm (measured along the pipe axis) from the end of the pipe in each case.
- (xi) The external surface of the plain ended pipes intended for collar or mechanically coupling joints shall be similarly prepared. Plain pipe ends for use with mechanical coupling joints shall conform to the requirements of the mechanical joint manufacturer. For large pipes where man access is possible, welding collars shall be provided for jointing cut pipes or closure pieces by means of internal and external fillet welding. The minimum length of each collar shall be 250 mm and the collar thickness shall be 1.5 mm greater than the equivalent standard straight pipe thickness. Welding collars may be provided as single split collars with temporary bolts and lugs. Split collar ends shall be prepared for butt welding in the same manner as plain ends of pipes prepared for butt welding. Collars shall make close contact around the circumferences of both pipes connected and the gap between the pipe ends in a split collar after tightening shall not exceed 3 mm. The tolerances of the collar shall be such that nowhere shall the gap between the inside surface of the collar and the outside surface of the pipe at fillet weld locations exceed the tolerances permitted for spigot and socket joints. The overlap on the welded collar on each pipe shall not be less than 75 mm. Collar joints shall not be required to take any deflection.
- (xii) All welded sleeve and collar joints shall be provided with two tapped holes spaced at 90 degrees apart on the circumference. They shall be of not less than 6 mm diameter and shall be provided with matching plugs to facilitate the air pressure testing of field welds. These tapped holes shall be located within 30 mm from the ends of the sockets and shall be cleared of welding runs.
- (xiii) Ends prepared for butt welding shall be subject to manufacturing tolerances in accordance with the relevant sections of API Standard 5L or Table 6.2 of API Standard 5LS.

- (xiv) Plain ends for use with mechanical joints or flange adaptor joints shall be truly circular with a diameter tolerance of ± 1 mm over a distance of 225 mm from the pipe ends.
 - (xv) Pipes to be used for closing lengths shall be correctly sized over their full length so that accurate alignment for split collar joints can be obtained. All such pipes shall be clearly marked.
 - (xvi) The joints for uPVC, polythene GRP and HDPE pipes shall be made using mechanical type or solvent welding or thermofusion techniques. The method of jointing shall be to the approval of the S.O.. The pipes shall be laid and jointed strictly in accordance with the manufacturer's instructions and shall conform to BS 5955-7.
- (b) Installation Procedure
- (i) Before making any joint the Contractor shall ensure that the interior of each pipe or fitting is clean and that it remains clean. Immediately before starting a joint the Contractor shall clean the end of each pipe to be jointed and shall prepare the ends for jointing as necessary. All mechanical joints shall be cleaned and have their coating made good before assembly in accordance with the coating manufacturer's recommendations.
 - (ii) The Contractor shall use only the proper jointing parts as specified and obtained from the suppliers of pipes or fittings. All joints shall be assembled, tightened and made only on pipes which comply dimensionally and have been prepared in accordance with the manufacturer's instructions. All joints shall be capable of passing the test for individual joints and for the completed pipeline as specified.
 - (iii) Detachable flexible couplings shall be installed where detailed on the 3D model or Drawings. The joints shall be assembled and tightened in accordance with the manufacturer's instructions. Where shown on the 3D model or Drawings, longitudinal movement of pipes joined by mechanical couplings shall be restricted by a steel work harness approved by the S.O.. After installation the harness shall be cleaned and painted with two coats of bituminous paint before wrapping.
 - (iv) Where a change of direction is to be achieved by the deflection of a flexible joint of any kind as shown on the 3D model or Drawings or as required by the S.O., such deflection shall not exceed 50 % of the design value for the size and type of joint as recommended by the manufacturer.
 - (v) Bolts for flanged or other mechanical joints shall show two threads clear of the nut after tightening.
 - (vi) Stainless steel nuts and bolts of grade A4-70 according to BS EN ISO 3506 (unless otherwise specified) shall be lubricated before assembly with an anti-galling lubricant. Nuts shall be tightened to the torques recommended by the joint manufacturer. Torques shall not exceed values to avoid the risk of galling.
 - (vii) After each joint has been completed, any protective or other coating shall be made good and any lining shall be completed as specified or detailed on the 3D model or Drawings without delay. Any metal joint which is not already coated shall be cleaned and painted with two coats of bituminous paint to BS 3416 (excluding Clause 10).

(c) Spigot and Socket Joints

- (i) In jointing iron spigot and socket pipes and specials with flexible joints, the Contractor shall take account of the manufacturer's recommendations as to the methods and equipment to be used in assembling the joints. In particular, the Contractor shall ensure that the spigot end of the pipe to be jointed is smooth and has been properly chamfered, that the rubber ring (with or without teeth insert type) is correctly positioned in the socket and that the two pipes are accurately in line before the joint is made. After the joint has been made the Contractor shall ensure by the use of a suitable feeler gauge that the rubber ring (with or without teeth insert type) is evenly seated in its correct final position. The rubber rings and any recommended lubricant shall be obtained only through the pipe supplier.

(d) Flanged joints

(i) General

- 1) All flanges shall be of the same or compatible material as the pipe or fittings, welded by the electric arc process or other approved method. They shall conform in all respects with the requirements of BS EN 1092 unless otherwise specified. They shall be of the raised face type or, where thermoplastic pipe materials require, flat face type and shall be truly faced over their whole width. Bolt holes shall be drilled off centre lines, truly in line end to end with the longitudinal axis. In general, flanges shall be rated to not less than PN 10 unless otherwise specified. All flanges shall be adequate to withstand test pressures for the fittings to which they are attached.
- 2) The requirement for bolt holes to be drilled off-centre must also be true for flanges on equipment to which the pipe flanges are connected.
- 3) All materials required for use in the making of flanged joints including nuts, bolts, washers and joint gaskets shall be supplied by the Contractor. Joint gaskets shall be contained within the bolt pitch circle and shall be made from 3.0 mm to 4.5 mm thick rubber to BS 1154 Class Y3 reinforced with two layers of fabric in accordance with BS EN 751. Each bolt shall be supplied and installed with a nut and two washers (one on either side of the flanges to be joined) and each bolt shall be of sufficient length to show two threads past the nut when so installed.
- 4) Test certificates for the flange material shall be provided to the S.O.. The finish on flange joint surfaces shall be in accordance with BS EN 1092.
- 5) Blank flanges shall be designed and supplied by the Contractor for an end loading equivalent to the rating pressure of the flange. Lifting eyes or handles and air release cocks shall be provided as necessary.
- 6) Thrust flanges shall be welded to the outside surface of pipework and shall be designed to withstand a longitudinal force equal to the loading applied to a blank flange of equivalent diameter.
- 7) The Contractor shall be responsible for checking and ensuring that mating flanges are compatible in all cases, including where connections are to be made to existing pipe flanges and to equipment to be provided as part of the Works.

(ii) Installation

- 1) When assembling flanged joints, the gasket may be fastened to the bolts with cotton thread. The use of jointing paste, adhesive tape or grease will not be permitted. The bores of abutting pipes or fittings shall be concentric with the joint gasket positioned centrally; no gasket material is to be left protruding into the bore. All nuts shall first be tightened by hand and nuts on opposite sides of the joint circumference followed by alternately and progressively tightening with a spanner so as to ensure even pressure all-round the joint. Care shall be taken to see that the gaskets remain in position undistorted while the bolts are tightened. For precise tightening, pre-set torque wrench shall be used.
- 2) Flanged joints located in chambers shall be painted with two coats of bituminous epoxy paint but flanges which will be subsequently buried in trench shall not be encased in concrete. These shall also be carefully wrapped with cold applied anti-corrosion and sealing tape such as 'Denso' tape or similar material approved by the S.O.. Buried ductile iron pipes shall be exteriorly coated with polyurethane coating. Where flanged pipes are to be buried for special reasons to be made known to the S.O., the pipe manufacturers' safe working bending moment values shall not be exceeded.

(e) Mechanical Couplings, Flange Adapters and Expansion Joints

(i) General

- 1) Mechanical couplings for jointing plain ended pipes shall be of an approved make, capable of maintaining a watertight joint over a range of axial movement between the pipe ends of at least 80 mm and with up to 3 degrees angular deflection between the longitudinal axes of the pipes.
- 2) Where mechanical couplings are used to take up the settlement in structures, they shall be provided in pairs.
- 3) The body of the mechanical coupling shall be manufactured from either ductile iron or mild steel. The material of construction shall comply with the following standards as a minimum:
 - a) Ductile Iron to BS EN 1563 EN-GJS-450-10 or ASTM A-536, Grade 65-45-12
 - b) Mild Steel to: BS EN 10025-2: Grade S275.
- 4) The complete coupling, comprising housing, gasket and fasteners, shall be the product of a single manufacturer.
- 5) Flange adapters for jointing plain ended pipes to fittings shall be of an approved make capable of maintaining a watertight joint over a range of axial movement of at least 25 mm and an angular deflection of not less than one and a half degrees.
- 6) Expansion joints shall be provided to conform with details as shown on the 3D model or drawings. They shall be supplied complete with all accessories.

- 7) Mechanical couplings, flange adapters and expansion joints shall be provided with tie rods wherever necessary. Pipe supports shall not be used to absorb thrust. Where tie rods are necessary, the rods shall be of the same diameter as the flange bolts. Flange adapters which require tie rods shall be of suitable dimensions to allow the tie rods not to interfere with the body of the flange adapter. Flange adaptors shall be tied to pipework with appropriate length of rods at each quadrant of pipe and not to pipework supports. Cutting or grinding of flange adapters to allow installation of tie rods shall not be permitted.
 - 8) All necessary couplers, gaskets, nuts, bolts and washers, etc. required for completing joints shall be supplied by the Contractor. All bolts, nuts and washers shall be stainless steel 316 as minimum unless otherwise specified.
 - 9) Couplings shall be supplied with or without a central register or locating plugs as required. The central collar shall be at least 1.5 mm thicker than the equivalent standard pipe thickness.
 - 10) Every coupling, flange adapter and expansion joint shall be capable of withstanding without leakage the pressure required for the works hydraulic test of the section of pipeline in which it will be incorporated. The pressure rating shall be clearly stamped on all couplings and adapters. The rubber joint rings shall be Type 1 to BS EN 681 PARTS 1 AND 2 having a hardness range of 45 to 65 degrees measured in accordance with BS 903 and tensile stress strain properties detailed in Table 3 corresponding to the relevant hardness.
 - 11) All metallic parts are to be shot blasted to SA 2½ to BS 7079 and protected with two coats of approved two pack epoxy based coating System 22 as elaborated in the Painting and Protective Coatings general specification unless otherwise stated. The enamel shall be taint free, non-toxic and chlorine resistant.
 - 12) All welding protrusion shall be machined finished. Mechanical couplings and flange adaptors shall be hydraulically tested at the place of manufacture one in every five for each size of coupling and adaptor.
 - 13) Every expansion joint shall be hydraulically tested at the place of manufacture prior to delivery.
- (ii) Installation of Flexible Mechanical Couplings for Iron or Steel Pipes
- 1) Mechanical couplings and flange adaptors joints shall be installed in strict accordance with the instructions given by the manufacturer and to the S.O.'s approval. In particular, the Contractor shall render the end of each pipe perfectly smooth so as to allow the joint sleeve to slide freely and where necessary shall re-coat the pipe ends with two coats of quick drying bituminous solution.

- 2) Before jointing, the ends of each pipe shall be wire brushed to remove any protective material adhering to the coating, together with any protuberances which may have been caused by rubbing of the packing material. Further, all rust at exposed places due to stripping of the bitumastic coating shall be removed. The ends shall then be recoated with a quick drying solution so as to produce a smooth coat of uniform thickness.
 - 3) On completion of the joint the surfaces of the coupling and adaptor shall be painted with a quick drying primer and an external sheathing mould formed to enclose the coupling and ends of the adjoining pipes. This work shall be carried out by specially competent and skilled men in strict accordance with the instructions given by the manufacturer.
 - 4) The primer and compatible sheathing material on the pipes together with the mould boxes and funnels required to form the protective mould and whiting used for lining the mould boxes prior to each pour shall be supplied by the Contractor, who shall also supply the necessary boilers to heat the sheathing material to the application temperature. The Contract Rates for making coupling and adaptor joints shall cover all costs in complying fully with the requirements of this Clause.
 - 5) Mechanical couplings and flange adaptors located in chambers shall be painted with two coats of approved bituminous epoxy enamel unless otherwise specified.
- (iii) Gaskets For Flexible Joints
- 1) Gaskets for joints shall be obtained from the manufacturer of the pipe or joint.
 - 2) Joint gaskets shall comply with the relevant requirements of BS EN 681 PARTS 1 AND 2, shall be of a material suitable for making a long term flexible seal between the pipes and shall be resistant to mechanical, chemical or bacteriological attack leading to deterioration of the flexible seal. Joint gaskets shall be entirely compatible with the media being conveyed by the pipeline.
 - 3) Lubricants used during jointing shall have no deleterious effects on the joint rings, pipes or pipe coating.
- (iv) Mechanical grooved couplings:
- 1) Mechanical grooved couplings shall consist of two segments. The coupling shall be complete with elastomer gaskets conforming to ASTM F1476 or BS EN 681 PARTS 1 AND 2, suitable for the intended service. Couplings using more than two segments housing are not acceptable.
 - 2) A written warranty shall be issued by the coupling manufacturer that the mechanical grooved coupling including gasket is designed for a service life not less than 20 years. In addition, couplings from the manufacture shall have successful installation track record not less than 10 years' service.
 - 3) The groove shall be created in compliance with the manufacturers installation guidelines.

G4.2.16 Welding

- (a) The Contractor shall engage a full time experienced AWS Certified Welding Inspector (CWI) for the inspection of all field welding.
 - (i) The responsibilities of the CWI shall include:
 - 1) Verify conformance of use of specified materials and their proper storage;
 - 2) Monitor and report conformance with approved welding procedure specifications;
 - 3) Monitor and report conformance with approved non-destructive test procedure specifications;
 - 4) Monitor and report conformance with requirements for each welder's performance qualification;
 - 5) Provide 100 percent visual inspection before, during and after field welding;
 - 6) Supervise non-destructive testing personnel and evaluate test results;
 - 7) Maintain records and prepare reports confirming inspection and testing in a timely manner; and
 - 8) Notify S.O. of unsatisfactory weld performance within 24 hours of weld test failure.
 - (b) General
 - (i) Where the steel pipes are to be joined by welding it shall be by means of an internal and an external circumferential electric arc weld at each joint.
 - (ii) Welded joints in trench shall generally be of the parallel short sleeve, spherical spigot and socket type with a circumferential full strength fillet weld both internally and externally. Welding gap tolerances shall be as specified. Full strength fillet welds shall have a throat thickness not less than 0.7 times the thickness of the pipe to be welded.
 - (iii) Where pre-jointing of straight runs of pipe outside the trench is permitted by the S.O., full penetration butt welds executed by manual or automatic procedures to the approval of the S.O. may be adopted. Butt welds shall have a throat thickness not less than the thickness of the pipe wall. During welding, the pipe ends shall be securely held by an approved method ensuring a correct and uniform root gap.
 - (iv) Before placing the pipes together the portion of the sockets and spigots to be welded shall be cleaned to a bright metallic finish, the spigot end shall then be placed in the socket and forced inside so that the spherical surfaces are in contact and that the gap between the pipes at the end of the spigot is nowhere greater than 1.5 mm. For longitudinally welded pipes, the longitudinal welds on adjacent pipes shall be at least 15° out of line. Details of the socket and spigot welded joint are shown on the drawings of the relevant code.

- (v) Where pipes are to be joined by a welded collar joint, the pipe ends and the collar shall be cleaned to a bright metallic finish. The collar shall then be placed centrally over the two ends to be jointed so that the gap between the pipe ends shall be about 25 mm from the centre line of the collar. The end of each pipe shall be welded to the sleeve collar using circumferential full strength fillet welds both internally and externally. Where a split collar is used, the two halves shall be drawn together by tie bolts until the gap between them is less than 3 mm. The four circumferential welds shall then be made, after which the tie bolts shall be removed and the longitudinal seam between the collar pieces welded. The tie bolt lugs shall then be cut off and the attachment points smoothed before the welds are tested.
- (vi) All parts to be welded shall have scale, slag, loose rust, paint and other foreign matter removed by means of a mechanical brush and shall be left clean and dry immediately before welding. All scale and slag shall be removed from each weld run when it is completed before welding over any previously deposited weld metal and prior to air testing.
- (vii) The electric generators used for welding shall be specifically manufactured for welding and shall have such controls and regulators as are necessary to adjust and control the voltage and current of the arc so that the conditions under which the electrode works shall conform to those recommended by the makers of the electrode. Electrodes shall be supplied as necessary to suit the different conditions of working. All equipment shall conform to the regulations of the Relevant Authority or of any other authority recognised by the S.O..
- (viii) The Contractor shall comply with the SS 510: 2005 "Code of Practice for Safety in Welding and Cutting (and other operations Involving the use of heat)" issued by the Ministry of Manpower. The Contractor is required to adopt the stipulated colour coding of gas hoses and to ensure that welding and cutting operations at the worksite conform to the Code of Practice.
- (ix) The Contractor shall note that no welding is allowed at any location where any existing gas transmission pipe is exposed.
- (x) All gas cutting sets shall be fitted with hose check valves. These are non-return valves fitted to the blowpipe ends of the oxygen and fuel hoses to arrest the flame at the inlet connections of the blowpipe, should a flashback occur during welding or cutting operations. All welding tools are to be kept free from oil and grease and main cylinder valves shall be shut-off when carrying out maintenance works on welding equipment.
- (xi) Field welded joints shall be either double welded butt strap joints, welded butt joints or double welded lap (spherical socket-spigot) joints and shall conform to approved welding procedures and referenced welding codes.
- (xii) Preheat and interpass temperature requirements for base metals shall be determined according to AWS D1.1 or other approved standards.

- (xiii) When fitting up the ends of the pipe to be welded or fitting butt-strap pieces or a canted collar, minor jacking or clamping will be allowed. Cold working the metal with sledges or localised application of heat and working the metal with sledges will not be allowed. For field displacement of joints where butt strap joints are required and which does not allow fit up with tolerances, special closure butt straps or mitred pieces shall be shop fabricated and installed.
- (xiv) Welded Lap Joints: During installation of welded steel pipe in either a straight or curve alignment, the pipe shall be laid so that there is a minimum lap at any point around the circumference of the joint as specified in the Contract Documents. The toe of the weld shall also be held back from the nearest point of tangency of the bell radius as shown in the drawings. Prior to welding, the pipe shall be shimmed at the joints to equalise the gap between the bell and the spigot around the joint circumference. The welding shall be performed in a manner that will maintain the equalised fit up.
- (xv) Butt-Joint Welds: Provide complete joint penetration welds at all butt-joint welds.
- (xvi) Lap joint welds for pipes larger than 700 mm shall be fillet welded on the inside and outside of the pipes.
- (xvii) Prior to the beginning of the welding procedure, any tack welds or joint stops used to position the pipe during laying shall be removed. Any annular space between the faying surfaces of the bell and spigot shall be equally distributed around the circumference of the joint by shimming, jacking, or other suitable means. The weld can then be made in accordance with the approved procedure.
- (xviii) Electrodes
 - 1) All electrodes shall be suitable for use on their respective base metals. Unless otherwise specified, all electrodes used shall be of low hydrogen type AWS A5.1 E7016/E7018 or equivalent suitable for pressured pipelines. All electrodes used shall be suitable for welding of mild steel containing not more than 0.26% carbon and shall comply with the requirements of BS 639: "Specification for Covered Electrodes for the Manual Metal-Arc Welding of Carbon and Carbon-manganese Steel"; other standards that are equal or superior to the British Standard may be accepted, subject to the S.O.'s approval.
 - 2) The type and particulars of the electrodes to be used on the work of the Contract having been determined, no change of type shall be made throughout the Contract period without the prior approval of the S.O..
- (xix) All welds shall be subject to inspection by the S.O. and shall comply with the requirements of Clause 8.9 of BS EN 1011-1 for freedom from undercutting, fusion penetration and soundness.

- (xx) After completion of welding at each joint, visual inspection, and non-destructive testing shall be carried out accordingly. The Contractor shall make available all the necessary arrangement apparatus and equipment and a specialist to carry out the test. All such tests shall be carried out by the Contractor in the presence of the S.O.. Where butt joints are being welded mechanically, the S.O. may permit the proportion of weld radiographically examined to be reduced if and when and as long as he is satisfied with the results being obtained. The standard of acceptability of welds shall be in accordance with API STD 1104 Section 6. Where necessary, welds shall be repaired in accordance with Annexes C and D of API 5L. Weld repairs shall not be made more than once at any position on the joint.
- (xxi) The analytical test reports in duplicate shall be submitted to the S.O. three days after the test. The repaired welds shall be radiographed again at no extra cost to the Board.
- (c) Weld on Offtakes
 - (i) For small offtakes below 100 mm diameter where no compensating plate is used, the fillet weld made externally shall be so formed that each leg of the weld shall not be less than one and a half times the thickness of the steel plate forming the branch pipe.
 - (ii) For offtakes or branches of 100 mm diameter and above fitted with compensating plates, fillet welds shall be formed internally as well as externally. In such a case, each leg of the fillet weld shall be not less than the thickness of the plate forming the branch offtake.
- (d) Examination before Welding
 - (i) Verify size, material, joint types, elevation, horizontal location, and pipe service of existing pipelines to be connected to new pipelines or new equipment;
 - (ii) Inspect size and location of structure penetrations to verify adequacy of wall cast-in pipes, sleeves, and other openings; and
 - (iii) Welding Electrodes: Verify proper grade and type, free of moisture and dampness, and coating is undamaged.
- (e) Preparation before Welding
 - (i) Notify S.O. at least 2 weeks prior to field fabrication of pipe or fittings;
 - (ii) Inspect pipe and fittings before installation, clean ends thoroughly, and remove foreign matter and dirt from inside; and
 - (iii) Damaged Coatings and Linings: Repair using original coating and lining materials in accordance with manufacturer's instructions and recommendations, except for damaged glass-lined pipe or PVDF-lined pipe that is to be promptly removed from the Site.
- (f) Mock-up Welding
 - (i) Mock-up welding shall be carried out if specified or requested by S.O. prior the work commencement on site.

- (ii) Mock-up welding shall be carried out separately on the mock-up pipe and fittings to simulate the actual welding of the special fabricated saddle onto the existing pipeline.
 - (iii) Full time third party independent welding inspector shall be engaged for inspection of this mock-up welding.
 - (iv) Mock-up weld shall be subjected to visual inspection and non-destructive testing as agreed by S.O. with the Contractor.
 - (v) Welding Procedure Specifications, and inspection and test reports of the mock-up welding shall be submitted to the S.O.. The actual site work can only proceed after the welding inspector is satisfied with the mock-up test results and the results are accepted by the S.O..
 - (vi) If the mock-up weld joint fails to pass the inspection by the welding inspector, the Contractor shall rectify the problem and re-test or propose an alternative welding method for the S.O.'s consideration.
- (g) Welding
- (i) Perform in accordance with Section IX, ASME Boiler and Pressure Vessel Code and ASME B31.1 or B31.3 for Pressure Piping, as may be specified on Piping Data Sheets, and if recommended by piping or fitting manufacturer.
 - (ii) Weld Identification: Mark each weld with symbol identifying welder.
 - (iii) Pipe End Preparation
 - 1) Machine Shaping: Preferred;
 - 2) Oxygen or Arc Cutting: Smooth to touch, true, and slag removal by chipping or grinding; and
 - 3) Bevelled Ends for Butt Welding: ANSI B16.25.
 - (iv) Surfaces Preparation
 - 1) Clean and free of paint, oil, rust, scale, slag, or other material detrimental to welding.
 - 2) Clean stainless steel joints with stainless steel wire brushes or stainless steel wool prior to welding.
 - 3) Thoroughly clean each layer of deposited weld metal, including final pass, prior to deposition of each additional layer of weld metal with a power-driven wire brush. Use stainless steel wire brush on stainless steel joints.
 - 4) Where stainless steel has been contaminated with carbon steel by the utilisation of carbon steel tools, chemically clean (passivate) exterior surfaces of piping to remove the contamination.
 - (v) Alignment and Spacing
 - 1) Align ends to be joined within existing commercial tolerances on diameters, wall thicknesses, and out-of-roundness;
 - 2) Root Opening of Joint: As stated in qualified welding procedure; and
 - 3) Minimum Spacing of Circumferential Butt Welds: Minimum four times pipe wall thickness or 25 mm, whichever is greater.

- (vi) Climatic Conditions: Do not perform welding if there is impingement of any rain or high wind on the weld area.
 - (vii) Tack Welds: Performed by qualified welder using same procedure as for completed weld, made with electrode similar or equivalent to electrode to be used for first weld pass, and not defective. Remove those not meeting requirements prior to commencing welding procedures.
 - (viii) Surface Defects: Chip or grind out those affecting soundness of weld.
 - (ix) Weld Passes: As required in welding procedure.
 - (x) Weld Quality: Free of cracks, incomplete penetration, weld undercutting, excessive weld reinforcement, porosity slag inclusions, and other defects in excess of limits shown in applicable piping code and shall have tensile strength across the weld not less than that of the thinner of the connected sections, and shall be water-tight.
- (h) Welding Procedure
- (i) Welding of joints in trench for steel pipes shall be carried out manually by the metal-arc process complying generally with the requirements of BS 4515.
 - (ii) Before starting the welding of pipe joints on Site, the Contractor shall submit for the S.O.'s approval details of the plant, methods and materials he proposes to use, including the following details:
 - 1) Make, type and gauge of electrodes;
 - 2) Size, shape and number of runs in the welded joints;
 - 3) Direction of welding;
 - 4) Current strength;
 - 5) Arrangements for air testing; and
 - 6) Welders' certificates.
 - (iii) Test specimens on pipes of the same size and thickness as the pipelines to be welded shall be made. These pipes shall be supplied by the Contractor. The joints shall be tested in the presence of the S.O. for each procedure proposed in accordance with the requirements of Clause 8 of BS 4515.
 - (iv) Only procedures approved in writing by the S.O. shall be adopted in the welding of the pipelines, and change from one procedure to another will not be permitted without submitting the new procedure for retesting and review.
- (i) Welders and Welder Performance Tests
- (i) Only the most highly skilled welders shall be employed on the Works, and every welder before commencing any joint welding shall prepare specimens for testing in accordance with Clause 7 of BS 4515 for each welding procedure proposed by the Contractor. Every welder shall be tested in accordance to Clauses 8 and 9 of BS 4515. The Contractor shall be responsible for the preparation, setting up, provision of specimen, all necessary equipment and materials for the testing of welding personnel.

- (ii) The Contractor shall inform the S.O. in writing which welders pass the performance qualification test. Welding shall be carried out only by welders approved by the S.O. and each welder shall identify his own work by means of a stencilled mark.
 - (iii) The Contractor shall maintain an up-to-date list of welders approved by the S.O. and, if ordered by the S.O., he shall remove from the approved list any welder whose workmanship, as demonstrated by the results of air pressure and/or other tests and visual inspection of individual welded joints, is in the S.O.'s opinion below the required standard.
- (j) Weld Inspection Criteria
- (i) S.O. and Contractor shall agree on the number and selection of welds to be tested.
 - (ii) Unless otherwise specified, perform NDT of welds at a spot testing frequency as shown below in accordance with the referenced welding codes as follows. In case there is a conflict the higher frequency level of NDT shall apply:
 - 1) For Pipework
 - Butt Joint Welds: 10 percent randomly radiographically tested and repaired;
 - Groove Welds: 10 percent randomly ultrasonically tested and repaired;
 - Fillet Welds: 10 percent randomly examined and repaired, using either dye penetrant or magnetic particle inspection methods; and
 - All Welds: 100 percent visually inspected.
 - 2) For Pipe Support
 - Butt Weld Joint: 100 percent radiographically tested and repaired;
 - Penetration Welds: 100 percent ultrasonically tested and repaired;
 - Fillet Welds: 100 percent examined and repaired, employing the magnetic particle inspection method using the half-wave rectified direct and alternating current; and
 - All Welds: 100 percent visually inspected.
 - (iii) Weld Acceptance:
 - 1) Visual Inspection (VT):
 - Perform 100 percent VT of all welds;
 - Structural Tubing: AWS D1.1, paragraph 6.9, Visual Inspection, Tubular Connections;
 - All Other Structural Steel: AWS D1.1, paragraph 6.9, Visual Inspection, Statically Loaded No tubular Connections; and
 - Stud Connections: AWS D1.1, paragraph 7.8.1.
 - 2) Ultrasonic Testing (UT): Perform UT of groove welds in accordance with AWS D1.1, paragraph 6.13.3, Class R Indications.

- 3) Radiographic Testing (RT): Perform RT of butt joint welds in accordance with AWS D1.1, paragraph 6.12.1.
 - 4) Magnetic Particle (MT):
 - Perform on fillet and partial penetration groove welds in accordance with AWS D1.1, paragraph 6.10; and
 - Acceptance shall be in accordance with VT standards specified above.
 - 5) Liquid Penetrant (PT):
 - Perform on fillet and partial penetration groove welds per AWS D1.1, paragraph 6.10; and
 - Acceptance shall be in accordance with VT standards specified above.
- (k) Air Testing of Welded Joints
- (i) After each joint has been welded it shall be air tested in the presence of the S.O..
 - (ii) A tapped hole, not less than 6 mm diameter, shall be provided in the socket end of each pipe unless otherwise specified, into which a suitable non-return valve and pressure gauge shall be fitted for purposes of the test. An air pressure of 17 bar shall then be pumped into the annular space between the spigot and socket and the pump disconnected. In the case of split collar joints, a tapped hole shall be provided at each end of the collar and each side of the joint shall be pressurised.
 - (iii) If no drop in pressure occurs over the ensuing period of ten minutes, the test shall be deemed successful subject to careful visual inspection for leakage during the test. If the test pressure is not maintained for ten minutes, the position of the leak shall be determined using soap solution.
 - (iv) All defects in the weld shall be cut back and rewelded as directed by the S.O. and the test reapplied until successful.
 - (v) The tapped test holes shall be then sealed off with welding into position the threaded screw plug after each joint has been satisfactorily tested.
 - (vi) The Contractor shall provide all necessary gauges, non-return valves, isolation valves, pumps, etc., for the air testing.
 - (vii) The requirements of this clause shall not limit the Contractor's obligations in achieving satisfactory results in the tests on completion as prescribed later.
- (l) Weld Defect Repair
- (i) Repair and retest rejected weld defects until sound weld metal has been deposited in accordance with the appropriate codes and the following:
 - 1) Removed by grinding or approved method, without impairing parent metal. No gas cutting to remove defective weld is permitted; and
 - 2) Re-weld for approval.

- (m) Thermal Fusion Jointing Equipment
 - (i) All tooling and equipment used for thermal fusion jointing shall be of types in accordance with British Gas Standards or equivalent and shall include all equipment necessary to cut, plane and clean, clamp, heat and align pipe and indicate or control temperatures and heating and cooling in accordance with the Board/ British Gas Standards.
 - (ii) Fusion tools and heating plates shall be electrically heated and thermostatically controlled with any handles or brackets made of material having electrical insulating properties conforming to BS 2559 : Part 3 for use with 230 V A.C. or if made of wood they shall conform to BS 3823 : Part 1. The tool or plate shall be earthed through the generator.
 - (iii) A thermometer shall be fitted and shall have a green zone indicating the correct fusion temperature range for HDPE marked with the appropriate code. Thermometers shall be suitably protected from mechanical damage, shall be replaceable and shall be accurate to $\pm 3^{\circ}\text{C}$ within the fusion temperature range. If an alternative temperature indicating system is used, the plate shall have provision for inserting a suitable conventional thermometer.
 - (iv) The wattage of the heater elements shall be such that the required surface temperature range is reached within 15 minutes in still air and at ambient temperatures of $20^{\circ}\text{C} \pm 5^{\circ}\text{C}$.
 - (v) The input leads shall conform to BS EN 50525 or other agreed standards.
 - (vi) The temperature control system (thermostat) shall be flameproof in accordance with BS EN 60079.
 - (vii) Hydraulic rams used to align pipe ends and bring them together shall each have independent pressure gauges. Each gauge shall have a performance certificate indicating any deviations in true to actual readings issued by an independent tester not less than one month prior to its use. The certificates shall be made available to the S.O., together with data relating the hydraulic pressure of the machine and the total force applied to the pipe.
- (n) Welding of Duplex and Super Duplex Stainless Steel
 - (i) The requirements for welding duplex stainless steel (DSS) and super duplex stainless steel (SDSS) as stipulated in the American Society of Mechanical Engineers (ASME) B31.3 Process Piping.
 - (ii) American Petroleum Institute (API) TR (Technical Report) 938-C, Use of Duplex Stainless Steels in the Oil Refining Industry, should be consulted for welding considerations.
- (o) Welding Processes
 - (i) Permitted welding processes shall include the additional restrictions and limitations as specified herein.

- (ii) When using the Submerged Arc Welding (SAW) process, the flux specified in the Welding Procedure Specification (WPS) is restricted to the specific flux brand name used in the welding procedure qualification test. The manufacturer and flux brand name shall be recorded on the Procedure Qualification Record (PQR) and the Welding Procedure Specification (WPS). Any change in the flux brand name shall require a new welding procedure qualification. Only flux trade brand names specified by the manufacturer as suitable for welding with DSS or SDSS filler metals are permitted. The electrode diameter size shall be limited to 3/32 inch (2.4 mm) maximum.
- (iii) For the Shielded Metal Arc Welding (SMAW) and Flux Cored Arc Welding (FCAW) processes, the electrode manufacturer and brand name of the consumable shall be the same for production welding as was used for welding the procedure qualification test coupon.
- (iv) For all welding processes, the maximum pass thickness shall not exceed 1/4 inch (6 mm); this limitation shall be specified in the WPS.
- (p) Fabrication Control
 - (i) The ferrite content of deposited weld metal (WM) for non-sour service applications shall be within the range suitable for the intended application under the governing environment.
 - (ii) Rough ground surfaces on the process side shall be final finished with a 100 grit or finer, flexible-backed abrasive (e.g., sanding disk, flap drum, flap wheel, etc.).
 - (iii) Abrasives used for grinding and sanding operations shall be manufactured free of iron, sulphur, and chlorine.
 - (iv) Heat tint darker than yellow-straw colour is not allowed. Heat tint darker than yellow-straw colour shall be removed by a method approved by the Board/ S.O.
 - (v) Welding purge shall be maintained, as necessary, to minimize the formation of process-side heat tint.
 - (vi) Filler metal addition during Gas Tungsten Arc Welding (GTAW) welding is required. Filler metal should be added at a minimum rate of 1 inch (25 mm) of filler deposited per 1 inch (25 mm) of weld length.
 - (vii) Prior to production welding of DSS or SDSS, the responsible contractor shall ensure that all fitters, welders, welding operators, and inspectors are aware of and adhere to the following restrictions.
 - (viii) Manufacturer shall perform inspections and maintain written records of the inspections, as necessary, prior to fabrication, during fabrication, during welding, and after welding to ensure that materials and workmanship meet the requirements of the design documents.
 - (ix) The following should be noted during the inspection:
 - 1) Examination
 - Verify size, material, joint types, elevation, horizontal location, and pipe service of existing pipelines to be connected to new pipelines or new equipment;

- Inspect size and location of structure penetrations to verify adequacy of wall cast-in pipes, sleeves, and other openings; and
 - Welding Electrodes: Verify proper grade and type, free of moisture and dampness, and coating is undamaged.
- 2) Preparation
- Notify S.O. at least 2 weeks prior to field fabrication of pipe or fittings;
 - Inspect pipe and fittings before installation, clean ends thoroughly, and remove foreign matter and dirt from inside; and
 - Damaged Coatings and Linings: Repair using original coating and lining materials in accordance with manufacturer's instructions and recommendations, except for damaged glass-lined pipe or PVDF-lined pipe that is to be promptly removed from the Site.
- 3) Welding
- Perform in accordance with Section IX, ASME Boiler and Pressure Vessel Code and ASME B31.1 or B31.3 for Pressure Piping, as may be specified on Piping Data Sheets, and if recommended by piping or fitting manufacturer;
 - Only approved welding procedures are used. Maximum interpass temperature and welding heat input limitations specified in the WPS are strictly adhered to. Filler metals are correct for the application and in accordance with the WPS; these metals shall be properly identified, stored, and issued and shall meet the required S.O.'s requirements specified herein;
 - Only properly trained and qualified welders are used;
 - Weld Identification: Mark each weld with symbol identifying welder; and
 - Pipe End Preparation:
 - Machine Shaping: Preferred;
 - Oxygen or Arc Cutting: Smooth to touch, true, and slag removal by chipping or grinding; and
 - Bevelled Ends for Butt Welding: ANSI B16.25.
- 4) Surfaces
- Clean and free of paint, oil, rust, scale, slag, or other material detrimental to welding.
 - Clean stainless steel joints with stainless steel wire brushes or stainless steel wool prior to welding.
 - Thoroughly clean each layer of deposited weld metal, including final pass, prior to deposition of each additional layer of weld metal with a power-driven wire brush. Use stainless steel wire brush on stainless steel joints.

- Where stainless steel has been contaminated with carbon steel by the utilisation of carbon steel tools, chemical clean (passivate) exterior surfaces of piping to remove the contamination.
- 5) Alignment and Spacing
- Align ends to be joined within existing commercial tolerances on diameters, wall thicknesses, and out-of-roundness;
 - Joint Design: As stated in qualified welding procedure;
 - Root Opening of Joint: As stated in qualified welding procedure; and
 - Minimum Spacing of Circumferential Butt Welds: Minimum four times pipe wall thickness or 25 mm, whichever is greater.
- 6) Climatic Conditions: Do not perform welding if there is impingement of any rain or high wind on the weld area.
- 7) Tack Welds: Performed by qualified welder using same procedure as for completed weld, made with electrode similar or equivalent to electrode to be used for first weld pass, and not defective. Remove those not meeting requirements prior to commencing welding procedures.
- 8) Surface Defects: Chip or grind out those affecting soundness of weld.
- 9) Weld Passes: As required in welding procedure.
- 10) Weld Quality: Free of cracks, incomplete penetration, weld undercutting, excessive weld reinforcement, porosity slag inclusions, and other defects in excess of limits shown in applicable piping code.
- 11) Paint sticks/temperature indicating crayons, antisplatter compounds, tapes, etc., have a total halogen content of less than 200 parts per million (ppm).
- 12) Backing gas (purge) is used for single-welded, complete joint penetration (CJP) joints, and purging is performed in accordance with the WPS and the purging procedure.
- (x) Joint Design
- 1) Root spacing for single-welded, CJP designs shall be in accordance with the following requirements:
- 3/32 inch (2.4 mm) minimum to 5/32 inch (4 mm) maximum for base material thickness exceeding 3/16 inch (5 mm); and
 - 1/16 inch (1.5 mm) minimum to 1/8 inch (3 mm) maximum for base material thickness 3/16 inch (5 mm) and less.
- 2) Root spacing for double-welded, CJP joint designs shall be limited to 5/32 inch (4 mm) maximum unless the final side of the joint to be welded is the surface exposed to the corrosion source in service.

(xi) Welding Consumables

- 1) The following consumables shall be used unless otherwise authorized by S.O.. The following welding consumable selection table does not address all DSS and SDSS alloys. The fabricator may propose 25 percent Cr consumables for welding 22 percent Cr base materials to S.O. for approval, especially if necessary to pass any specified corrosion-resistance testing requirements.
- 2) Consideration should be given to purchasing filler metals within a predicted ferrite number (FN) range that is within the range required for production weld deposits. For example, the WRC-1992 diagram may be used to estimate the final weld deposit FN using the actual chemical composition for each heat/lot of filler metal. For the determination, nitrogen content must be included in the analysis. A rough approximation for converting FN to volume percent ferrite is $FN \times 0.7 = \text{percent ferrite}$.

Base Material		Bare Wire (ASME Specification SFA 5.9)	SMAW (ASME Specification SFA 5.4)	FCAW Electrodes (ASME Specification SFA 5.22)
Type	UNS No.	American Welding Society (AWS) Classification	AWS Classification	AWS Classification
2304	S32304	ER2209	E2209	E2209TX-X
2205	S31803/S32205	ER2209	E2209	E2209TX-X
255	S32550	ER2594	E2594	E2594
2507	S32750	ER2594	E2594	2594 Equivalent

- 3) To join DSS to SDSS, either filler metal specified for welding the respective similar base material combinations may be used to weld the dissimilar materials, with the S.O.'s approval.
- 4) Refer to Articles Q215.6.1 and Q215.7 for specific S.O.'s requirements for 25 percent Cr filler metals.

5) Selected consumables shall meet the requirements in the following table:

Welding Process	Specification	Classification ⁽¹⁾	Notes
GMAW, GTAW, SAW	SFA-5.9	ER2594	(1)
GMAW, GTAW, SAW	BS EN ISO 14343	2594 N L	(1)(2)(3)
SMAW	SFA-5.4	E2594-XX or E2595-XX	(1)
	BS EN ISO 3581	E2594 N L	(1)(3)(4)
FCAW	(5)	(5)	(1)
Notes: - The pitting resistance equivalent (PRE) number shall be calculated for each heat and lot of Type 2594 consumable used in production, using the actual chemical composition from the certified material test report (CMTR) and utilizing either of the following formulas. Values are expressed in weight percent. The minimum PRE value shall meet or exceed 40.0. Formulas are as follows: $PRE_N = Cr + 3.3Mo + 16N$ $PRE_W = Cr + 3.3Mo + 1.65W + 16N$ - Optional product form designations: G = GMAW filler metal, W = GTAW filler metal, S = SAW electrode (e.g., GTAW filler material specification/classification designation, EN 12072 W 2594 N L). "2594 N L" filler material classification designation corresponds to the "25 10 4 N L" classification designation by some filler material manufacturers. - Optional supplementary S.O. designators defined in EN 1600 (e.g., type of covering [R = rutile, B = basic], percent WM recovery, and welding position) may be used. Basic electrodes are preferred for fabrications with Charpy impact toughness requirements. - Currently, there are no national or international specifications for Type 2594 FCAW filler materials. Therefore, the chemical composition of the filler material to be purchased shall meet Specification SFA A5.4 for Classification E2594 or E2595-XX or Specification BS EN 12072 and 2594 N L Classification. Any change in the trade name of the FCAW electrode shall require a new welding procedure qualification.			

(xii) Welding Interpass Temperature

- 1) The maximum interpass temperature shall be in accordance with the following table:

Base Material	Base Metal (BM) Nominal Thickness	Maximum Interpass Temperature
Duplex, e.g., 2205 (UNS S32205) (UNS S31803)	All	350° F (175° C)
Super Duplex, e.g., 2507 (UNS S32750) 255 (UNS S32550)	≤ 1/4 inch (6.4 mm)	200° F (95° C)
	> 1/4 inch (6.4 mm)	300° F (150° C)

(xiii) Purge and Shielding Gas

- 1) Backing/purge gas is required to be argon, nitrogen, or argon/nitrogen blends; and
- 2) Hydrogen additions for torch gas shielding or backing/purge gas in any amount are not permitted.

(xiv) Duplex and Super Duplex Procedure Qualification Requirements (PQR)

The following Production Thickness:

For production thicknesses less than 1/2 inch (12.7 mm), the procedure qualification test coupon thickness shall not vary by more than ±20 percent of the production thickness.

1) Single-Welded, Complete Joint Penetration Groove Welds

- Production welding of single-welded, CJP groove welds shall be qualified utilizing a single-welded, CJP test coupon (without back welding), with the same groove type for procedure qualification testing. The groove type used for procedure qualification testing shall conform to the applicable type to be used in production (e.g., single V-groove, single U-groove, or single bevel, etc.). The joint design for the qualification testing shall be within the manufacturer's design tolerances used for production welding.
- Procedure qualifications for single-welded, CJP welds shall individually list welding parameters and heat input used for welding the root pass, the second layer, and the balance of welding.
- WPSs utilized for single-welded, CJP welds shall individually list welding parameters and heat input for the root pass, the second layer, and the balance of welding. The listed welding parameters on the WPS shall be within appropriate ranges to be able to operate within the specified heat input ranges for the different weld layers.
- The WPS shall restrict the heat input during welding (for the root pass and hot pass layer) to the PQR heat input values used for the specific layers ±10 percent.

- Procedure qualifications on single-welded, CJP groove welds shall require protective gas backing.
 - A change in the composition of the backing/purge gas shall require a supplementary qualification test using the new backing gas during welding of the test coupon. The specimen shall be the same P-No. and Group No. as the production base material. The thickness of the specimen shall be sufficient to allow for at least three weld deposit layers. The same joint design and welding parameters for the three layers shall be the same as was used for the original PQR. Where three weld layers are not possible because of reduced thickness, the number of production weld passes shall be duplicated. A metallurgical section shall be removed from the test coupon, polished, and etched; the ferrite content shall be estimated in accordance with ASTM E562, Standard Test Method for Determining Volume Fraction by Systematic Manual Point Count. The evaluation shall be performed in the heat affected zone (HAZ) on each side of the weld in the region of the root pass (total of two) and the weld root. The acceptance criteria for ferrite content shall be in accordance with the section on Fabrication Control. Additionally, a photomicrograph(s) of the weld root, hot pass, and HAZ (200X minimum magnification) shall be included with the PQR and provided to S.O. for review.
- 2) PQR Filler Metal Documentation
- SMAW and FCAW welding consumable manufacturer(s) and brand name(s) shall be reported on the PQR;
 - For the Submerged Arc Welding (SAW) process, the flux manufacturer and brand designation shall be listed on the PQR; and
 - Manufacturers' certified material test reports (CMTRs)/certifications for base materials and welding consumables should be included with the PQR. The CMTRs should indicate the actual chemical composition, including nitrogen.
- 3) PQR FN Testing Requirements
- The FN shall be measured at three locations on the weld face surface and at three locations on the weld root surface of the procedure qualification test coupon. Spacing between readings should be at least 3 inches (0.76 mm). The acceptance criteria for the weld procedure qualification shall be 30 to 70 FN, as determined by an instrument calibrated to AWS A4.2.
- 4) PQR Volume Percent Ferrite Determinations
- A metallurgical section (or sections) removed from the PQR test coupon shall be suitably polished and etched to clearly reveal the two-phase austenite/ferrite microstructure (i.e., phase balance) of the WM, HAZ, and BM.

- A visual estimate of the ferrite volume percent content shall be made in accordance with ASTM E562, Standard Test Method for Determining Volume Fraction by Systematic Manual Point Count. The evaluation shall be performed in the HAZ on each side of the weld in the region of the root pass (total of two), the HAZ on each side of the weld in the region of the cap pass (total of two), and three measurements in the WM near the vertical centre line of the weld (one in the cap, one in the root, and one at mid-thickness).
 - For non-sour service applications, the acceptance criteria shall be 22 to 50 volume percent.
- 5) PQR Microstructural Evaluations
- Photomicrographs (at 200X minimum magnification) shall be included with the PQR and provided to the S.O. for each of the locations where a phase balance is performed. The microstructure is required to be substantially free of a continuous ferrite network in the HAZ. A continuous ferrite network is subject to be rejected.
- 6) PQR ASTM A923 Testing Requirements
- Test specimens from the PQR test coupon shall be evaluated in accordance with ASTM A923, Standard Test Methods for Detecting Detrimental Intermetallic Phase in Duplex Austenitic/Ferritic Stainless Steels. Test Methods "B" and "C" shall be required to be performed.
- 7) ASTM A923, Method "B," Charpy Testing Requirements
- For Test Method "B," the quantity and location of Charpy specimens shall be in accordance with the following requirements:

Weld Joint Thickness	Notch Location				
	Weld Center Line	Fusion Line (FL)	FL +0.10 in. (2.5 mm)	FL +0.19 in. (4.75 mm)	Notes
≤ 1 in. (25 mm)	2	2	2	2	1
> 1 in.	2	2	2	2	2
Notes: 1. The specimen orientation shall be across the weld, with the notch axis normal to the material surface with one face of the specimen ≤1/16 inch (1.6 mm) from the material's surface that corresponds with the weld root. 2. The specimen orientation shall be across the weld, with the notch axis normal to the material surface with one face of one of the specimens ≤1/16 inch (1.6 mm) from the material's surface that corresponds with the weld root and one face of the other specimen ≤1/16 inch (1.6 mm) from the material's surface that corresponds with the weld face.					

Impact testing of Charpy specimens shall be performed at -40° F (-40° C).

Acceptance criteria for full-size 0.39 inch x 0.39 inch (10 mm x 10 mm) specimens shall be in accordance with the following requirements:

WM: 25 ft-lb (34 Nm)

FL: 40 ft-lb (54 Nm)

FL + 0.10 inch (2.5 mm): 40 ft-lb (54 Nm)

FL + 0.19 inch (4.75 mm): 40 ft-lb (54 Nm)

The energy acceptance criteria for a sub size specimen will be reduced in direct proportion to the reduced area of the sub size specimen relative to that of the full-size specimen.

(xv) ASTM A923, Method "C," Ferric Chloride Corrosion Requirements

- 1) For Test Method "C," testing and acceptance criteria shall be in accordance with the requirements of ASTM A923, Method "C," with the following modifications:
 - Full thickness specimens shall include at least 0.5 inch (12 mm) base material on each side of the weld toe, and the specimen in the welding direction shall be a minimum dimension of 2 inches (50 mm);
 - 22 Cr base materials and weldments (regardless of weld deposit composition) shall be tested at 22° C (72° F); and
 - 25 Cr base materials and weldments (regardless of weld deposit composition) shall be tested at 40° C (104° F).
- 2) In addition to the requirements of ASTM A923, visual examination for pitting shall be performed at 10X minimum magnification. Evidence of pitting shall be measured and documented in accordance with ASTM G48, Standard Test Methods for Pitting and Crevice Corrosion Resistance of Stainless Steels and Related Alloys by Use of Ferric Chloride Solution, Clause 14. When pitting is observed, the sides of the test specimen exhibiting pitting shall be photographed at 2X minimum magnification.

(xvi) PQR Hardness Testing

- 1) A hardness test specimen (cross section) from the procedure qualification test coupon shall be removed from the qualification test coupon and suitably ground, polished, and etched to distinguish the base material, HAZs, and WM. Hardness testing shall be performed in accordance with the following requirements:
 - Hardness surveys for non-sour service applications shall be conducted using the Vickers (HV) 10 kg load. The surveys shall encompass the BM, HAZ, and weld deposit at the top and bottom of the weldment. A minimum of two traverses are required to be performed across the weldment cross section. Each traverse shall include a set of readings in a line approximately 0.06 inch (1.6 mm) from each base material surface, with at least one BM reading on each side of the joint, three HAZ readings (two readings if the HAZ is less than 0.12 inch [3 mm] wide) on each side of the joint (spaced approximately 0.04 inch [1 mm] apart with the first reading within approximately 0.02 inch [0.5 mm] from the weld fusion line), and two WM readings.

- Hardness restrictions for non-sour applications for 22 percent Cr welds shall not exceed the equivalent of 310 HV (average), and no individual reading shall exceed 320 HV.
- Hardness restrictions for non-sour applications for 25 percent Cr welds shall not exceed the equivalent of 330 HV (average), and no individual reading shall exceed 350 HV.

(xvii) Heat Input Controls

- 1) Heat input control for welding shall be specified in the applicable WPS. Heat input shall be determined by the following formula or other method as provided in ASME Section IX, QW-409.1:

$$\text{Heat Input [J/in. (J/mm)]} = \frac{\text{Voltage} \times \text{Amperage} \times 60}{\text{Travel Speed [in./min (mm/min)]}}$$

(xviii) Minimum Heat Input

- 1) The minimum required heat input shall be 12.5 kJ/inch (0.5 kJ/mm). If maintaining this minimum heat input is not possible due to material thickness or welding process limitations, prior approval is required by S.O..

(xix) Maximum Heat Input

- 1) The maximum heat input is dependent on the weld joint design and shall be limited to the following:

Single-Welded, Complete Joint Penetration Welds	
Root Pass	35 kJ/inch (1.4 kJ/mm) maximum heat input. Approximately 1/8 inch (3 mm) root bead thickness.
Second Pass	30 kJ/inch (1.2 kJ/mm) maximum heat input. Approximately 3/32 inch (2.4 mm) bead thickness
Remainder of Welding	45 kJ/inch (1.8 kJ/mm) maximum heat input
Additionally, for single-welded, CJP welds, the WPS shall restrict the heat input during welding (for the root pass and hot pass layer) to the PQR heat input values used for the specific layers +10 percent.	

Other Welds (Such as Fillet Welds, Partial Penetration Welds, or Double V-Groove Welds)	
All Passes	45 kJ/inch (1.8 kJ/mm) maximum heat input

- (xx) Postweld Heat Treatment
 - 1) Postweld heat treatment shall not be performed without prior written approval from S.O..
- (xxi) Production Non Destructive Examination (NDE)
 - 1) A minimum of 10 percent of completed weld joints and 2 inches (50 mm) on both sides of the weld joint shall be randomly selected for wet/dry exposure by misting the area with clean, potable water and examining the area for evidence of rusting after 24 hours. Note: This method is not always effective for revealing surface contamination. The fabricator or contractor shall be responsible for cleaning iron-contaminated surfaces whether or not the contaminated surface is exposed to the service process fluid.
- (xxii) Ferrite Measurements for Non-Sour Service
 - 1) For non-sour service applications, production ferrite determinations shall be performed on 10 percent of welds for each pipe size and wall thickness of completed butt and branch connection welds. The welds to be tested shall be randomly selected. Each weld test location shall consist of a minimum of three closely spaced tests (e.g., 0.5 to 2 inches [10 to 50 mm] spacing). The three readings in each general test locations may be averaged.
 - 2) For non-sour service applications, the acceptance criteria shall be within 30 to 70 FN.
 - 3) A rough approximation for converting FN to volume percent ferrite is $FN \times 0.7 = \text{percent ferrite}$.
 - 4) If any averaged values for ferrite tests are unacceptable, additional tests are required and shall be performed in the same area as close as practical to the failed test. If the average of the retests is within the allowable range, no additional testing is required. If the results of the additional tests are unacceptable, a non-conformance report documenting the BM, filler metal type, location, test results, and proposed disposition shall be submitted to S.O. for review. For every weld with unacceptable ferrite test results, an additional three welds not previously tested shall be tested. All test results shall be documented and traceable to the weld joint that was tested.

G4.2.17 Access to Interior of Pipelines

- (a) Pipes provided with cable pass holes shall be laid so as to provide pass holes positioned on top of the pipelines in excess of 600 mm in diameter at intervals of approximately 60 m. When a pass hole is no longer required, the screw plug shall be replaced and welding done to the pipe externally and the internal and external protection systems shall be made good over the plug and holiday tested.
- (b) The Contractor shall record the location of all cable hole passes and ensure the correct sealing thereof.

G4.2.18 Installation of Underground Pipes

(a) General

- (i) Trench excavation and backfilling shall be coordinated with the construction of the pipeline as a whole so as to ensure expeditious completion of the whole operation.
- (ii) Trench excavation and backfill shall be carried out in accordance with the specification for Earthworks and, where not otherwise specified, in accordance with BS EN 14161: 2011 + A1: 2015 and PD 8010-1: 2015 + A1: 2016. The minimum depth of cover above pipelines shall normally be 1000 mm, the depth of cover being measured from original ground level to the top of the barrel of the pipe. Any pipeline with less than normal cover shall be surrounded with at least 150 mm of concrete unless otherwise instructed.
- (iii) The lines and levels of the pipeline and the positions of the bends, junctions and other fittings which are shown on the 3D model or Drawings may be adjusted by the S.O. as necessary before construction of a particular section commences. The pipeline shall be accurately installed to the lines, levels, grades and positions set out by the Contractor from data supplied to him by the S.O. so that the barrel of the pipe shall be continuously supported throughout its length by the bottom of the trench or by bedding material as specified.
- (iv) The Contractor shall submit full details of the methods he proposes to use for control of the accuracy of pipelaying. Sight rails shall be used to control the accuracy of pipelaying unless the S.O. directs otherwise. Where sight rails are used, they shall be fixed and maintained at each change of gradient, and at as many intermediate points as may be necessary, but not more than 35 m apart. Sight-rails shall be clearly painted in contrasting colours and be not less than 1 metre long and 150 mm deep, straight and level, rigidly supported by stout wooden posts at each end. Any pipes placed on end for affixing the base of the posts shall be not less than 225 mm in diameter and shall be filled and rammed solid with earth or sand. Posts shall otherwise have their bases concreted in, to the approval of the S.O.. Boning rods or travellers for use with sight rails shall be of robust construction, clearly painted and accurately made to the various lengths required, the lower ends being provided with shoes with sufficient projection to rest on the pipe inverts.
- (v) Pipes shall be laid to even grades for as long a length as possible, with a minimum grade of 0.2 %, irrespective of local changes of elevation of the ground surface. The position of the internal face of any pipeline shall not deviate from that described in the Contract by more than plus or minus 20 mm. Notwithstanding, no pipe shall be laid with a backfill upon completion of the pipeline. Changes in direction or in grade of the pipeline shall be carried out by making use of any permissible deflection of joints between straight pipes or by the introduction of bends.
- (vi) Where the gradient exceeds 5%, installation shall proceed on an ascending grade with sockets leading. Unless specified otherwise pipes shall be installed singly and shall not be jointed until after they have been laid. After laying and jointing, the invert level of each pipe shall be checked before the next pipe is installed.

- (vii) Joint holes shall be excavated to provide adequate clearance to enable joints to be made satisfactorily and any specified wrapping to be completed and to provide a clearance of at least 50 mm from the trench bottom below the joints.
 - (viii) The Contractor shall keep the interior of pipes clean and free from water, dirt, stones and other foreign matter as installation proceeds, and at the end of the day's work. At other times when installation work is not proceeding the open ends of the pipes shall be sealed off by a suitable stopper. The Contractor shall take such precautions as are necessary to prevent pipes from floating.
 - (ix) Wherever pipes, cables, ducts and similar services are buried in the ground a marker tape shall be laid 300 mm above each and every service in the trench.
 - (x) Where gravity pipelines are to be bedded on concrete, precast concrete bedding blocks shall be set to a level allowing for the thickness of the pipe barrel, and side string lines shall be used for checking alignment.
- (b) Steel Pipes
- (i) No pipe shall be rolled into place for lowering into the trench. If a crane is used for handling the pipes into the trench, the pipes shall be slung with slings and lifting belts passed around the outside unless otherwise specified.
 - (ii) A recess shall be formed in the trench foundation beneath to permit each sling or belt to be withdrawn without damage to the coating or sheathing. All struts at the ends of the pipes and pipe specials shall not be removed until they are ready for engagement.
 - (iii) Before any pipe or special is lowered into the trench, it shall be cleaned and re-examined for cracks and flaws whilst hanging in the sling. If undamaged it shall be placed in position ready for jointing. Except for pipework to be embedded in concrete, all pipes and specials shall be placed on a solid and even foundation for their full length. Pipes and specials shall be placed in position singly and the order of laying pipes shall be subject to the approval of the S.O..
 - (iv) After laying, the interior of pipes or specials shall be carefully cleaned again to remove any debris, dirt, stones or other matter that may have entered during laying.
- (c) Installation of High Density Polyethylene (HDPE) Pipes
- (i) Installation shall be in accordance with the recommendations of BS EN 12201, WIS 4-32-08 and of the manufacturers of pipes, fittings, tooling and equipment. Should any of these requirements conflict, the Contractor shall inform the S.O. of the conflict and propose the procedure to be adopted with sufficient time to allow the S.O. to approve or modify the Contractor's proposals.
 - (ii) Thermal fusion jointing, whether by socket fusion, butt fusion or the use of electrofusion couplings, shall be carried out only by workmen who have been trained in such work by a manufacturer of HDPE pipes meeting this specification. The Contractor shall submit the names of thermal fusion jointers to the S.O. for approval together with full details and evidence of the relevant training and experience each has received. Only approved jointers shall be allowed to make each type of joint.

- (iii) Where not otherwise specified, pipes shall be jointed using electrofusion couplings or socket or butt thermal fusion. However, socket or butt thermal fusion shall be used only where the Contractor has made proposals for its use and for routine testing of a proportion of joints which have obtained approval from the S.O.. Should the S.O. become dissatisfied with the standard of socket or butt thermal fusion jointing, approval for its use may be withdrawn, in which case electrofusion couplings shall be used.
- (iv) Gradual changes in direction or grade of the pipeline may be made by bending the pipe provided that the pipe centre-line radius of the bend curvature shall not be less than 40 times the pipe nominal diameter.
- (v) HDPE pipe or fittings to be built into concrete shall first be wrapped in not less than 3 layers of heavy gauge polyethylene sheet.
- (d) Installation of Solvent Welded Thermoplastics Pipes
 - (i) Installation shall be in accordance with the manufacturer's recommendations. Solvent cements and cleaners are highly flammable and shall be stored in accordance with safe practices for such substances.
- (e) Bedding to Pipes
 - (i) Pipes for gravity pipelines shall be laid on bedding as detailed in the 3D model or Drawings. For pressure cement pipelines, and pressure and gravity steel pipelines, grey iron and ductile iron pipes, the bottom of the trench shall be finished smooth and free from irregularities, so that the pipes bear uniformly and are supported throughout their entire length. The holes required at each joint shall be made as small as possible.
 - (ii) Where instructed by the S.O. pressure pipelines of asbestos cement-pipes, and pressure and gravity pipelines of steel, grey iron and ductile iron pipes, shall be bedded on soft material from nearby trench excavations. The bedding material shall be placed and compacted using a vibrating plate or similar type of compacting machine, and shaped to provide continuous support to the bottom of each pipe along its full length.
 - (iii) Where the base of the trench excavation is in rock, a further 150 mm of rock shall be excavated below the intended bedding line for the pipe barrel and this excavation shall be refilled with soft material, or with sand free from stones greater than 19 mm size, or with concrete.
- (f) Concrete Bedding and Surround
 - (i) Where specified or indicated on the 3D model or Drawings or ordered by the S.O., pipes shall be bedded haunched or surrounded in concrete in accordance with the typical details shown on the 3D model or Drawings. Concrete shall be un-reinforced of the grade shown on the Drawings or itemised in the Bill of Quantities. Any formwork required shall be of Class F1.

- (ii) Where indicated on the 3D model or Drawings or ordered by the S.O., pipes shall be laid on a concrete bed with a minimum thickness under the pipe as shown on the 3D model or Drawings. Such concrete shall not be placed until the joints at each end of the pipe have been completed. The full width and depth of bedding concrete shall be placed and carefully punned beneath the pipe, followed at once by the addition of any haunching and surround concrete. To ensure uniform support under the pipe the bedding concrete shall be placed from one side of the pipe only and observed to flow under the pipe to the other side. Unformed surfaces shall be of spade finish. The pipe shall be prevented from floating or other movement during concreting.
 - (iii) Wherever pipes are provided with concrete protection they shall be supported temporarily on precast concrete blocks cast to the shape of the bedding and where they have a concrete surround the depth of the precast concrete block shall be increased to suit. Where pipes other than asbestos cement pipes are used the precast concrete blocks shall be positioned behind each socket. In the case of asbestos cement pipes a precast concrete block shall be positioned at each end of each pipe clear of the sleeve joints. The width of the precast concrete blocks shall be half the diameter of the supported pipe subject to a minimum and maximum of 115 mm and 300 mm respectively. The blocks shall be left in and paid for as in-situ concrete.
 - (iv) The continuity of the concrete bed, haunch or surround to pipes with flexible spigot and socket joints shall be broken completely by a 12 mm thick board of softwood, 'Flexcell' or other approved material cut to shape and placed against the face of the socket. The concrete bed, haunch or surround shall be discontinued at detachable flexible couplings leaving a distance of 150 mm at each side of the coupling.
 - (v) Where two or more pipelines are laid in the same trench the joints shall coincide at the points where the continuity of the concrete bed, haunch or surround is broken, i.e. at the joints in the pipeline with the longest pipes. Any intermediate joints in the other pipelines with shorter pipes shall be surrounded in concrete.
- (g) Granular Bedding and Surround
- (i) Where indicated on the 3D model or Drawings or ordered by the S.O., pipes shall be bedded with granular material up to the half pipe diameter and for the full width of the trench. uPVC and other 'flexible' pipes shall be surrounded with granular material across the full width of the trench and to a level at least 100 mm above the top of the pipe.
 - (ii) Granular bedding shall be well compacted to the correct levels so that pipes are given full support along the entire barrel length. Recesses shall be formed in the granular bedding to accommodate sockets, collars or other pipe joints. Pipes shall not be supported on timber wedges, pre-cast concrete blocks or other point supports.
 - (iii) After pipes have been correctly laid and tested further granular material for haunching or surround shall be placed and carefully compacted so no disturbance of the pipe occurs.

- (iv) Material for granular bedding and surround shall be broken stone or irregular shaped gravel and shall be inert to the ground water.
- (v) Unless otherwise specified, it shall have the following grading:

BS Sieve Size:	Percentage by mass passing sieve size
63 mm:	100
37.5 mm:	85 - 100
20 mm:	0 - 20
10 mm:	0 – 5
- (vi) The material shall be tested in accordance with BS EN 932 and BS EN 1097-2, using specimens in the soaked condition, and shall have a 10 % fines value greater than 50 kN. Material retained on a 7 mm sieve shall have an index of flakiness less than 25 % and an index of elongation less than 45 %.
- (vii) Whenever the S.O. directs, a test for compaction shall be carried out on the material for granular bedding in accordance with BS 5955.

G4.2.19 Protection

- (a) General
 - (i) The main process pipes i.e. water pipes shall be protected both internally and externally against corrosion. Where not otherwise stated herein this specification or the Particular Specification, or on the 3D model or the Drawings, the pipes shall be protected externally with a double reinforced bitumen enamel wrapping. Pipe interior shall also be protected with spun concrete lining. Both types of protection shall be applied under factory conditions but not necessarily at the place of manufacture of the pipes.
 - (ii) Internal and external protection at joints shall be completed on Site after acceptance of joint weld tests.
 - (iii) Pipes other than main process pipes shall be protected in accordance with the general specification section on Painting and Protective Coatings.
- (b) Cleaning and Priming
 - (i) Where bitumen wrapping is to be applied at the place of pipe manufacture, cleaning and priming shall immediately follow hydraulic testing. Where bitumen wrapping is to be applied at another location, the pipes shall be coated with a suitable corrosion inhibitor before despatch from their place of manufacture.
 - (ii) Immediately prior to priming, all pipes shall be thoroughly descaled by shot blasting internally and externally to the second quality standard according to BS 7079 and BS EN ISO 8501-1.
 - (iii) If the spun concrete lining is to be applied at the place of pipe manufacture, internal priming of the pipe barrel may be omitted. Otherwise pipes shall be primed both internally and externally.

- (iv) The above priming shall stop short of the ends of pipes intended for field welding. The ends of such pipes shall be coated with an approved priming paint and/or rust inhibitor both internally and externally. This shall be suitable for providing a temporary protection from rusting prior to field welding those areas of pipe not given the full factory applied permanent protection and shall not affect the weldability of the metal.
- (c) External Protection
 - (i) Bitumen Enamel Wrapping
 - 1) Pipes shall be coated with bitumen in accordance with BS EN 545 and BS EN 598 except that the protection shall have a minimum thickness of 6 mm for pipes over 324 mm OD. The bitumen shall be Type 2 of BS EN 10300 and there shall be not less than 2 mm of bitumen between the inner and outer wraps, and between the pipe and the inner wrap.
 - 2) The coating shall be stopped short 100 mm from ends prepared for butt welding, 150 mm from ends of all spigot and socket pipes and 250 mm from the ends of all plain ended pipes for use with mechanical couplings or flange adaptors. The edge of the wrapping shall be chamfered at 25 degrees to the axis of the pipe.
 - 3) The coated surfaces shall be protected with a suitably approved ensis oil or similar material during manufacture so that extensive cleaning of the surface is not required before and after jointing on site.
 - 4) The wrapping materials shall be spirally wound onto the pipes and fittings simultaneously with the bitumen coating. Each wrap shall be from 150 to 225 mm wide and the edges shall overlap by 12 to 25 mm. Care shall be taken to ensure that the inner wrap does not come into contact with the pipe metal or with the outer wrap.
 - 5) The inner wrap shall be a glass fibre resin bonded tissue reinforced in the longitudinal direction with parallel glass threads spaced 10 mm apart. The nominal thickness shall be 0.5 mm and the minimum weight shall 0.046 kg/m².
 - 6) The outer wrap shall be of glass fibre resin bonded tissue reinforced in the longitudinal direction with parallel glass threads spaced 10 to 25 mm apart. It shall be impregnated with a material fully compatible with the bitumen coating to give a finished thickness of 0.75 mm.
 - (ii) Inspection of External Pipe Coating
 - 1) Before dispatch from the factory, the wrapping of all pipes shall be rigorously inspected for defects. Thickness shall be determined by a pit gauge, continuity with a holiday detector and coating quality by cutting out 75 mm square samples at the rate of one sample per 5 lengths of pipe manufactured.
 - 2) The whole coated surface area of all pipes and fittings shall be tested for pinholes or other invisible defects in the coating using an approved holiday detector at a potential of 14,000 volts.

- 3) After the wrapping has been tested it shall be given one coat of a non-stick light and heat reflecting paint.
 - 4) Any length on which the coating is, in the opinion of the S.O. poorly applied shall be cleaned to bare metal and recoated. Minor defects may be repaired by touching up. All repairs shall be checked for thickness and continuity.
- (iii) Painting Coated Pipes and Fittings
- 1) All coated pipes and fittings shall be given two coats of undiluted commercial "Matex" or similar approved vinyl acetate paint to reduce the risk of the coating becoming tacky.
 - Epoxy Coatings
 - Refer to Painting and Protective Coatings Section of the General Specification.
 - Low Friction Coating
 - Where necessary, the external surfaces of the pipe ends for use with mechanical couplings and flange adaptors shall be given, after blast cleaning to the second quality standard according to BS 7079 and BS EN ISO 8501-1, an approved system of low friction vinyl based protective coating. The coating shall be applied in accordance with the manufacturer's instructions. Where zinc or bituminous coatings adjoin the low friction coating, they shall overlap the low friction coating by 25 mm. The length of pipe barrel to be so protected by the low friction coating shall be 300 mm unless directed otherwise; and
 - After curing but before removal from the factory the coating shall be wrapped with tape suitable for protecting the coating from damage in transit to and on the Site. The tape shall be removed immediately prior to installation of the pipe or fitting so coated.
- (iv) Completion of External Protection at Joints
- 1) All pipe joints in steel shall be protected from corrosion as specified.
 - 2) No external protection shall be completed at joints until, where applicable, electrical continuity has been provided and until the joint has been successfully tested. No backfilling at joints shall take place until the S.O. has inspected and approved the external protection of the joints.
 - Bitumen Enamel Wrapping
 - Bare metal shall be thoroughly cleaned to a bright metallic finish and it shall be immediately coated with a primer solution. The primer shall be applied cold by brush;
 - As soon as the primer has set the sheathing shall be completed by running hot bitumen into a metal mould placed over the joint and overlapping the sheathing by 75 mm on either side of the joint. The bitumen shall be according to BS EN 10300; and

- Priming and sheathing material for the purpose shall be provided by the Contractor including the supply of all the necessary mould boxes and funnels required to form the protective mould and whitening for use in lining the mould boxes prior to each pour. The Contractor shall also supply all necessary boilers to heat the bitumen to the application temperature recommended by the manufacturer.
 - Epoxy Coated
 - Epoxy coating shall be used to complete the external protection at joints and cable pass holes in factory epoxy coated steel pipe. The surface shall be prepared by blast cleaning to SA2½, primed, undercoated and painted with two coats of high build cured two pack epoxy coating as specified in the general specifications.
- (d) External Wrapping
- (i) Where specified or detailed on the 3D model or Contract Drawings, pipes, fittings and valves that are to be buried in the ground shall be protected with wrapping material. All items not wrapped before installation shall be clean and dry and shall be given one coat of primer supplied by the manufacturer of the wrapping material. Joint and fitting profiles shall then be modified by the application of moulding putty supplied by the manufacturer of the wrapping material to provide a suitable profile for wrapping. Wrapping shall then be carried out and shall lap with the works-applied protection on the pipes on each side of the joint. The wrapping shall be applied under tension to achieve conformability and intimate adhesion without trapped air pockets. Wrapping shall be carried out with a 55% overlap of adjacent strips. End overlaps where a new roll joins a completed one shall be 150 mm minimum.
 - (ii) Wrapping Material
 - 1) Wrapping material for external protection of buried pipelines and for site wrapping of completed joints shall be heavy duty, self-adhesive, rubber-bitumen compound with PVC carrier strip together with primer and moulding putty.
 - 2) All materials shall be supplied by the same manufacturer.
 - 3) The primer shall be a fast drying bitumen solution type compatible with the wrapping material and also with coal tar, bitumen enamel and epoxy coatings and with plastics pipes.
 - 4) The moulding putty shall be compatible with the wrapping materials and the primer and shall be capable of being moulded cold by hand.
 - 5) The wrapping material shall possess the following minimum physical characteristics:

Thickness (overall)	:	1.5 mm
Tensile strength	:	10 N/mm
Elongation	:	40%
Tear resistance	:	20 N force
Impact resistance	:	7 Joules
Adhesion (180 deg. peel)	:	2 N/mm

Dielectric strength:

(55% overlap)	:	15 000 V minimum
Insulation resistance	:	1 000 000 megOhms

- 6) The materials shall be suitable for application in the environmental conditions at the Site.

(e) Polyethylene Sleeving

- (i) Polyethylene sleeving for loose pipe wrap shall conform to BS 6076. Suitable adhesive PVC tape shall also be provided to fix the sleeving in position and to make joints between the adjacent sections.

(f) Fusion Bonded Epoxy Coating

- (i) Where shown on the 3D model or Drawings or otherwise stated, fusion bonded epoxy coatings shall be applied to ductile iron, cast iron, steel, and aluminium based substrates used in the manufacture of pipes, fittings and their components that will be in contact with potable water, domestic effluent, any soil conditions and above ground applications. Further details are given in General Specification section on Painting and Protective Coatings. The coating shall be factory applied at a factory approved by the S.O..
- (ii) Fusion-Bonded Epoxy: American Waterworks Association Standard (AWWA C213) for "Fusion Bonded Epoxy Coating for the Interior and Exterior of Steel Water Pipelines" or Australian Standard AS 3862.
- (iii) Epoxy used for lining of the pipes and fittings shall be fusion-bonded epoxy consisting of a one-component thermosetting epoxy that is cross-linked to the steel substrate. Material shall consist of epoxy resin, curing agents, catalysts, fillers, colorants, flow-control agents and ultraviolet light resistant agents.
- (iv) The internal lining shall have a total thickness of at least 0.5 mm for all sizes of pipes and fittings.
- (v) Material Properties
- 1) Shock resistance to be greater than 11 Nm;
 - 2) Elongation greater than 10 percent;
 - 3) The material shall show no effect when subjected to salt water spray test of 1000 hours;
 - 4) The material is to be chemically resistant to chlorinated water, water solution of salts, acids and bases within a pH range of 5 to 9;
 - 5) Cathodic disbondment: 8 mm radius;
 - 6) Adhesion to steel: 20,600 kPa; and
 - 7) Water absorption: Maximum 0.83 percent.

- (vi) Repair of fusion bonded epoxy coating
- 1) In the event of a fusion bonded epoxy coating becoming damaged after leaving the factory, the item shall be returned to the factory for repair or, with the consent of the S.O., be repaired at Site. Repairs shall be effected in accordance with the following requirements:
 - Repair materials shall be compatible with the original factory-applied coating materials;
 - The edges of the original factory-applied coating shall be ground off to a taper or feather edge;
 - Exposed metal shall be treated as specified in the General Specification section on Painting and Protective Coatings;
 - The relative humidity of the atmosphere in which the repair is to be effected shall be maintained (if necessary, by the use of hot air blowers and tenting) at less than 85%;
 - The surface temperature of the exposed metal shall be raised to at least 3 °C above dew point and the coating shall then be applied in stages, as recommended by the manufacturer, so as to achieve a total thickness in accordance with the General Specification section on Painting and Protective Coatings; and
 - All repairs shall be checked with a holiday detector and any holidays shall be repaired, after which the repairs shall be re-checked.
- (g) Rigid Polyurethane, 100 percent solids: American Waterworks Association AWWA C222, latest edition "Polyurethane Coatings for the Interior and Exterior of Steel Water Pipe and Fittings":
- (i) Polyurethane used for lining of the pipes and fittings shall be of 100 percent solids rigid polyurethane. The system shall be self-priming, plural component, suitable for immersion in chlorinated water.
 - (ii) The internal lining shall have a thickness of at least 1.0 mm for all sizes of pipes and fittings.
 - (iii) Material Properties:
 - 1) The dielectric resistance shall be 200 kV/cm;
 - 2) Density: 1600 kg/m³;
 - 3) The shock resistance greater than 12 Nm;
 - 4) Elongation greater than 10 percent;
 - 5) The material shall show no effect when subjected to a salt water spray test of 1000 hours;
 - 6) The material is to be chemically resistant to chlorinated water, water solution of salts, acids and bases within a pH range of 5 to 9;
 - 7) Cathodic disbondment: 12 mm radius;
 - 8) Adhesion to steel: 10,350 kPa; and
 - 9) Water absorption: Maximum 1.4 percent.

(h) Internal Protection

(i) Spun Concrete Lining at Factory

- 1) Main process pipework (clean water) application, carbon steel pipe including specials shall be lined with concrete in accordance with BS EN 10311 and BS EN 10224. Cement for lining shall be ordinary Portland cement to BS EN 197-1 or equivalent.
- 2) For used water application, carbon steel shall be lined with sulphate resisting cement (SRC).
- 3) Ductile iron pipe for used water application shall be lined with concrete in accordance with BS EN 598.
- 4) Aggregate shall be well graded clean fine aggregate and the maximum particle size shall not exceed 8 mm or one third the thickness of the lining, whichever is the lesser. All materials used in concrete for lining shall comply with the requirements for concrete.
- 5) The final aggregate grading and concrete mix proportions shall be such that a hard, durable and dense concrete lining is obtained that satisfies the tests laid down in BS EN 545 and BS EN 598. Unless otherwise approved, the minimum cement content shall be 330 kg/m³ and the water cement ratio of the mix loaded into the pipe shall not exceed 0.42. Tests shall be carried out during the lining of pipes to demonstrate that the concrete lining has a strength equal to or greater than the minimum figures stated in BS EN 10311. The frequency of these tests shall be at least once every 250 m length of pipe lined or during each working shift, whichever is the greater. The testing of concrete cubes shall be conducted at the factory of manufacture.
- 6) All straight pipes shall be concrete lined by the use of a lining machine designed and built for the purpose of rotating the pipe and centrifugally applying the lining at sufficient speeds to meet the requirements set out below. The support or holding device for the pipe shall be such as to avoid damage to the pipe coating during the spinning operation. If the pipe is rotated by direct contact drive from the machine it shall be supported over at least 90 degrees of its circumference and driven by non-metallic surfaced belts of sufficient width to avoid coating damage. The speed of rotation of the pipe during the lining compaction stage shall be such as to provide a radial acceleration of at least 250 m/s². The rotational drive shall be capable of close control and provide smooth acceleration and deceleration when working up to and down from the compaction spinning stage.
- 7) Immediately before lining commences, the pipe bore shall be cleaned of all loose scale, rust, oil, grease or any other foreign matter likely to contaminate or harm the concrete. Areas where an approved priming coat has bonded to the pipe such that wire brushing does not cause areas to flake off will be acceptable as a base for applying the lining.

- 8) The entire quantity of concrete required for the lining shall be applied without interruption, and spinning shall continue until the specified thickness is evenly distributed over the inside of the pipe, all surplus water has been removed and the greatest possible density of concrete has been obtained. Tolerance on the thickness of lining shall be as follows:
- +3 mm to 2 mm for 25 mm and above; and
 - +2 mm to 1 mm for 19 mm and below.
- 9) Temporary or semi-permanent end restraints shall be fixed to the pipe ends on completion of lining and before removal of the pipe from the lining machine. The pipe shall not be rotated about its longitudinal axis after the fixing of the restraints when being handled and transported to the storage yard and subsequently to the Site.
- 10) Fittings shall be concrete lined at the factory in such a manner that the lining shall be of the specified thickness and comparable in density and smoothness to the spun lining in straight pipes and generally applied in accordance with BS EN 10311 and BS EN 10224. During the lining process, all rebound material, dribbles, etc. shall be removed so that the lining is applied in a homogeneous mass to a clean surface.
- 11) Inspection of the lining shall be carried out at any time prior to and after installation of the pipes. Any pipe with lining that is broken, defective or otherwise not in accordance with this specification, the referred codes and standards shall be rejected. Remedial lining operations may be carried out if the method of remedial lining has been submitted in writing to the S.O. in advance and approved by the S.O.. The standard of the remedial lining shall satisfy the requirements of this specification.
- 12) Surface crazing of the lining will be acceptable unless cracks are severe enough such that they can be penetrated to a depth of 2 mm by a 250 microns feeler gauge at 10 points or more over a length of 300 mm when measured with the lining in a saturated condition. These cracks shall be cut back to full depth and sealed with an approval epoxy resin filler.
- 13) Concrete lining shall be omitted at the following locations:
- Pipe ends:
 - The lining shall terminate 100 mm back from both sleeve and sleeve-welded socket pipe ends. The edge of the lining shall be angled back at 75 degrees to the pipe axis in order to provide a positive key for in situ joint protection.
 - Plain ends:
 - For butt straps or collar joints, the lining shall terminate 100 mm back from the pipe end. The edge of the lining shall be angled back at 75 degrees to the pipe axis in order to provide a positive key for in situ joint protection; and
 - For mechanical coupling and flange adaptor joints, the lining shall be brought right against the retaining rings.

- 14) The unlined surfaces shall be protected with a suitably approved ensis oil or similar material during manufacture so that extensive cleaning of the surface is not required before and after jointing on site.
- Curing
 - Linings applied to pipes shall be cured in such a manner as will enable concrete to obtain and subsequently retain optimum strength, density and durability. Linings shall be kept moist by continuous water spraying for a period of at least 3 days. They shall then be protected from sunlight and kept damp by spraying with water or other means approved by the S.O. for a further period of 7 days. Pipes shall not be removed from the factory until at least 14 days have elapsed from the date of lining.
 - Epoxy Coatings
 - Refer to the General Specification section on Painting and Protective Coatings.
- (ii) Completion of Internal Protection at Joints
- 1) The Contractor shall not commence completion of internal protection at joints until the two pipes concerned have both been jointed and laid to the correct line and level in the trench and backfilled to the specified standard to at least 300 mm above the top of the pipe.
- Cement Mortar Lining
 - Hand placed in-situ mortar shall be used to complete the lining at joints and cable pass holes in factory cement mortar lined steel pipe. It shall be completed within 24 hours of backfilling that length of pipe;
 - Hand placed mortar shall be finished with a steel trowel and shall have a smooth surface of uniform appearance with even transitions to adjacent machine-placed lining;
 - Hand-placed in-situ mortar shall be in concrete as specified and shall consist of 1 part of cement to 1½ parts of sand by volume, or other approved mix, and shall be of suitable water content and consistency for adhesion and trowelling to a smooth finish. The thickness of hand-applied mortar shall be the same as the adjacent factory-applied lining. Where necessary wire mesh may be introduced for ease of completing thick concrete lining particularly those of spherical type joints;
 - Areas to which hand-applied mortar is to be applied shall be thoroughly cleaned of all loose material and grease and wire brushed. The areas shall then be moistened with water just prior to the application of mortar. Hand mortaring shall not commence until the Contractor has received written confirmation from the S.O. that the surface preparation is acceptable;

- Curing of the lining shall commence immediately after completion of the lining operation at each joint or area of defect. The section of the pipe to be cured shall be isolated and a moist atmosphere shall be maintained at the place to be cured to keep the mortar lining damp and to prevent evaporation of water from it. Curing shall continue for two weeks; and
 - Admixtures to improve the adhesion of the hand applied mortar to the pipe and curing compounds shall not be used without the approval of the S.O. and, if approved, shall be used in accordance with the manufacturer's recommendations or otherwise as determined by the S.O..
 - Epoxy Lining
 - Epoxy coating shall be used to complete the lining at joints and cable pass holes in factory applied epoxy lined steel pipe. It shall be completed within 24 hours of backfilling that length of pipe. The surface shall be prepared by blast cleaning to Sa2½, primed, undercoated and painted with two coats of high build cured two pack epoxy coating as specified in the General Specification section on Painting and Protective Coatings.
- (i) Protection of Pipes and Valves in Chambers
- 1) Factory applied coatings on pipes and valves in chambers and other exposed areas shall be made good on completion of construction of those chambers as necessary to return the items to the appearance on leaving the factory. Pipes and valves to be installed above ground or in chambers and tanks with a bituminous coating at the manufacturer's works shall be overpainted;
 - 2) Where a colour coded coating is specified or directed by the S.O., exposed pipes and valve; and
 - 3) Their joints shall be overpainted as follows:
 - Bituminous coated items shall be primed with two coats of aluminium sealer to a total dry film thickness not less than 30 microns. Top coats of approved colour of alkyd chlorinated rubber finish shall be applied sufficient to give a total dry film thickness of 80 microns;
 - Fusion bonded epoxy or epoxy coated items shall be lightly rubbed down with emery to provide a key and painted with two coats of alkyd or epoxy finish coats of approved colour; and
 - Paint systems for overpainting shall be compatible with any factory coatings including coatings made good on Site.

- (j) Painting Pipework above Ground
 - (i) Steel pipework for installation above ground shall be blast cleaned to Sa 2½ to BS 7079 Part A1 and coated with epoxy iron oxide two pack epoxy blast primer, 25 microns dry film thickness, and painted with 75 microns minimum dry film thickness of two pack polyamide cured epoxy zinc phosphate primer at the place of manufacture. After installation, damaged areas of paintwork and welded areas shall be power brush cleaned and painted with 75 microns dry film thickness of the same paint. Subsequently, this whole pipeworks shall be painted in accordance with the General Specification section on Painting and Protective Coatings.
 - (ii) All buried iron pipes and valves including flexible mechanical couplings shall be protected from corrosion with wrapping material as specified above.

G4.2.20 Pipe Supports

- (a) Hangers and brackets for supporting exposed pipes shall be fabricated from mild steel and shall be painted in accordance with the General Specification section on Painting and Protective Coatings.
- (b) Design of support systems shall allow for a minimum factor of safety of 2 on the ultimate tensile strength of the support material.
- (c) The Contractor shall submit details and samples of the proposed pipe support system for the approval of the S.O. prior to commencing procurement of material, fabrication or installation.
- (d) Workmanship - Handling
 - (i) General
 - 1) Throughout the construction of pipelines including the operations of off-loading, delivery, handling, storing and transporting pipes, valves and precast manhole sections on or about the Site, the Contractor shall use such methods, plant and equipment as will prevent damage to the pipes, valves, precast manhole sections and to any sheathing, lining or paintwork, and such methods shall include the use in appropriate cases of pipehooks, lifting beams, reinforced canvas slings, protective padding, struts, cradles and pipe trailers.
 - 2) Temporary packing, coverings or crates provided by the suppliers for the protection of pipes and valves in transit shall not be removed (except for purposes of inspection after which they shall be replaced) until immediately before the pipe or valve is installed, and shall then be disposed of by the Contractor.
 - 3) No pipe shall be moved by rolling.
 - (ii) Steel Pipes
 - 1) Great care shall be exercised to ensure that the pipes and specials are not subject to any kind of shock or sudden load during hauling and handling.
 - 2) Coated steel pipes shall be lifted and moved only by wide non-abrasive slings or belts or by other means acceptable to the S.O.. Wire ropes, chains and hooks shall not be permitted to come in contact with the coating.

- 3) The pipes shall be so handled, stored and transported as to prevent undue distortion and shall not be moved in any manner involving rotation of the pipe about the longitudinal pipe axis.
 - 4) The pipes shall be lifted by means of two reinforced canvas slings or belts at least 300 mm wide suspended from a lifting beam so that the slings are positioned at a distance of one fifth of the pipe length from each end of the pipe.
 - 5) The Contractor shall provide suitable timber end struts and sufficient intermediate struts to strengthen the pipes to the S.O.'s approval to prevent distortion during handling and delivery. Such struts shall be left in position when thin-walled pipes of 1400 mm and above are delivered to the storage areas.
- (iii) Pipe Supports for Exposed Pipework within Buildings
- 1) Exposed pipes within buildings shall be supported by approved stanchions, beams, channels, angle bars, clips, hangers or rollers at intervals not exceeding those given in the tables of maximum spacing between pipes supports at the end of this section.
 - 2) The supports shall permit appropriate movement for expansion and contraction in the desired direction and prevent transmission of vibration.
 - 3) Branches shall be supported so that they are free to move and so that undue strain is not thrown on the necks of valves or other fittings.
 - 4) Supports shall be arranged so that stresses or loads due to expansion or contraction, pressure reaction, and the weight of pipes and their contents, which are transmitted do not damage the building structure or the pipes themselves.
 - 5) Where piping of material other than steel is being supported by steel supports and the contact between dissimilar metals could cause electrolysis, then the pipework shall be isolated from the supports in a manner approved by the S.O..
 - 6) Where pipes run together in parallel, their supports shall be grouped.
 - 7) Fastening of supports to the building or structure shall be of means approved by the S.O. prior to any fixing being made. Wooden plugs and explosive powered fasteners shall not be used. Steel work shall not be drilled or welded to unless approval has been given by the S.O..
 - 8) Vertical uPVC pipes passing through floors shall be supported with uPVC collars attached to the pipes which shall allow for movement of the pipes during temperature changes.
 - 9) Pipes shall not be sprung into place but shall hang freely and shall not transmit vibration to the building structure.
 - 10) Vibration isolation supports shall comprise neoprene in shear and helical spring assemblies.

- 11) Anchors shall be provided where necessary to provide reactions for expansion devices, flexible type connections and to prevent excessive expansion or contraction forces in pipework from being transmitted through structure.
- 12) On steel pipes, anchors may be either welded or clamped securely to the bare pipe.
- 13) On copper pipes the design of anchor shall embody a sleeve welded to the bracket and clamped round the pipe with separation between dissimilar metals to prevent electrolysis.
- 14) Any pipe found to be "noisy" during testing due to insufficient support or loose fixing will be rejected.

MAXIMUM SPACING BETWEEN PIPE SUPPORTS FOR EXPOSED STEEL,
 DUCTILE IRON AND COPPER PIPES

	Steel Pipes	Ductile Pipes	Copper Pipes
Relevant Standard	BS EN 10226	BS EN 545 , BS EN 598	BS EN 1057

MAXIMUM SPACING BETWEEN PIPE SUPPORTS FOR uPVC PIPES

Nominal Size (mm)	Maximum Horizontal Support Spacing (mm)	Maximum Vertical Support Spacing (mm)
32.0	500	1200
40.0	950	1200
50.0	600	1200
65.0 and above	900	1800

G4.2.21 Inspection and Testing

- (a) Specialist Inspection And Testing
 - (i) The Board may, under a separate contract, employ a specialist firm to give the S.O. such assistance as he may require in any matter connected with pipeline materials, including the inspection of materials and workmanship and the witnessing of tests at any stage (including manufacture) during the execution and maintenance of the Works.
 - (ii) The S.O. may require the specialist firm to supervise welding procedure tests and welder performance tests and to carry out various independent tests such as holiday tests, Pearson tests, magnetic crack detection tests and radiographic examination of welded joints. Such independent inspection and witnessing of tests may be carried out at any stage during the execution and maintenance of the Works but shall not relieve the Contractor of any of his obligations under the contract.

- (iii) To the extent instructed by the S.O., the Contractor shall provide labour, plant and materials (but not special testing equipment) for direct assistance to the specialist firm in their inspection and independent testing and for any further work of investigation and repair which the S.O. considers necessary as a result of such inspection or testing. The provision of such labour, plant and materials will be paid for by the Board at fair and reasonable rates to be determined by the S.O., except in cases where in the S.O.'s opinion the inspection, test or further investigation shows that materials and workmanship provided by the Contractor do not comply with the specified requirements, whereby the cost shall be borne by the Contractor.
- (b) Inspection at Time of Installation
 - (i) In addition to any inspection and tests made when delivery is taken, pipeline materials, including any sheathing, lining or protective paintwork, shall be inspected by the Contractor immediately before and after installation and any damage shall be repaired by the Contractor as directed by the S.O. before the pipe or fitting is installed or jointed as the case may be. Any special material required for the repair of pipe sheathing or lining shall be obtained from the pipe supplier and shall be used in accordance with his recommendations.
 - (ii) The S.O. may himself, and without thereby relieving the Contractor of any of his obligations, inspect and test the pipeline materials by any means he considers appropriate and any damage discovered by such inspection shall be repaired by the Contractor.
 - (iii) The Contractor shall remove from the Site and shall provide a replacement for any pipe or fitting, which in the opinion of the S.O. is so badly damaged as to be unfit for repair on the Site.
- (c) Inspection Prior to Water Testing
 - (i) Prior to the filling of gravity or pressure pipelines with clean water for water testing, each pipeline shall be thoroughly inspected internally jointly by the Contractor and the S.O. to ensure that the line is cleaned from stones, dirt, debris, plastic paper or any other foreign matter. Any foreign matter present shall be removed prior to water testing.
 - (ii) Where the pipeline diameter is less than 600 mm such that man entry is not feasible, the Contractor shall demonstrate that the pipeline is thoroughly clean and free from foreign matter to the satisfaction of the S.O.. The means of demonstration may include flushing, swabbing, high pressure internal jetting and Remotely Operated Vehicle (ROV).
- (d) Cleaning Pipelines
 - (i) The pipelines shall be kept thoroughly cleaned and ready for inspection by the S.O. at any time and an adequate supply of expanding rubber plugs shall be available all times to ensure that all pipe-ends are stoppered off. Except when cleaning or water testing, water shall not be allowed to flow through the pipes.

- (ii) Where the pipeline diameter is less than 600 mm such that man entry is not feasible, gravity pipelines and ducts shall be proved by passing through them a train consisting of an approved badger and a circular rubber squeegee followed by a wooden ball unless otherwise directed. The wooden ball diameter shall be smaller than the diameter of the pipes under inspection by not more than 25 mm. The Contractor shall remove any obstruction blocking these operations.
- (iii) Before commencing pressure testing the Contractor shall clean out the part of the pipeline to be tested to the satisfaction of the S.O.. All materials and equipment necessary for cleaning shall be provided by the Contractor unless otherwise specified. Full records shall be kept of the cleaning and shall be submitted to the S.O. as soon as practicable after each cleaning operation.
- (iv) The method of cleaning shall be proposed by the Contractor and be subject to the approval of the S.O..
- (v) After the whole pipeline has been successfully pressure tested (whether as a whole or in sections) and the Contractor has removed all temporary works and reconnected any part temporarily removed from the pipeline, the Contractor shall finally clean out the whole pipeline and flush it through with clean water.
- (vi) Flushing water shall be discharged only to sites or to conduits that are approved by the S.O. for the discharge.
- (vii) Discharges which cause damage, create a nuisance or health hazard or interfere with the work of the Board or others shall not be permitted.
- (viii) The Contractor shall be liable to the Board for any damage caused to the pipeline or to pumps and other equipment of the Board as a result of foreign matter of any kind not having been cleared out of the pipeline before it is handed over to the Board.

G4.2.22 Testing Pipelines

(a) General

- (i) The Contractor shall subject all parts of the completed pipeline system to a final test. For the purposes of this final test, the pipeline system shall if necessary be divided into suitable sections as agreed with the S.O.. All testing shall be witnessed and certified as successful or unsuccessful by the S.O..
- (ii) In general, the test pressure shall be 1.5 times the maximum working pressure or the maximum working pressure plus 5 Bar whichever is lower or the surge pressure if this is greater.
- (iii) All steel pipelines shall be tested in accordance with BS EN 805 and PD 8010-1: 2015 + A1.
- (iv) GRP pipes shall be tested in accordance with BS EN 14364.
- (v) Ductile Iron pipes shall be tested in accordance with BS EN 545 for water pipelines and BS EN 598 for sewerage pipelines.
- (vi) uPVC pipes shall be tested in accordance with BS 3506.
- (vii) Polyethylene pipes shall be tested in accordance with the procedures in 'Pressure Testing of Water Supply Pipelines and Sewer Rising Mains' published by WRc PLC.

- (viii) The Contractor shall notify the S.O. at least five clear working days beforehand of his intention to test a section of pipeline.
- (ix) Pipe section shall be tested in sections no greater than 200 metres in length, unless otherwise approved by S.O..
- (x) Install temporary thrust blocking or other restraint as necessary to protect adjacent piping or equipment and make taps in piping prior to testing.
- (xi) Prior to test, remove or suitably isolate appurtenant instruments or devices that could be damaged by pressure testing. Isolate all equipment from the test. Tests against valves are not acceptable.
- (xii) New piping connected to pipes installed under other contracts: Isolate new piping with grooved-end pipe caps, spectacle blinds, blind flanges, or other means as acceptable to the S.O..
- (xiii) Items that do not require testing include: Piping between wetwells and wetwell isolation valves, equipment seal drains, tank overflows to atmospheric vented drains, and tank atmospheric vents.
- (xiv) Test section may be filled with water and allowed to stand under low pressure prior to testing.
- (xv) Install temporary pressure relief valve at the lowest point in the piping system, set at 7 kPa above system test pressure, to ensure that the system is not over pressurised. The S.O. shall be present during system pressurisation.
- (xvi) Pre-insulated steel pipe (PIP) insulation closure pieces shall not be installed until leakage testing has been conducted to the approval of the S.O..
- (xvii) Where directed by the S.O. a section of pipeline in a particularly difficult location shall be subjected to a hydrostatic test immediately when the section has been completed, using a pressure not less than that to be applied for the final test.
- (xviii) Notwithstanding the foregoing, the S.O. may at any stage of construction carry out such preliminary tests as he considers desirable to check materials and workmanship on the pipeline but this shall not relieve the Contractor of his obligations to achieve approved test results on completion.
- (xix) Weld testing and inspection and pipe deflection testing must be completed and approved before pressure testing will be allowed.
- (xx) The Contractor shall during testing demonstrate satisfactory operation of the concerned pipework under the working pressure of each valve and hydrant installed under the Contract.
- (xxi) The water required to carry out testing and subsequent sterilisation (if applicable) of all pipelines shall be deemed to have been included under the Contractor's scope of supply. Where situations are encountered in which special considerations render water unavailable or there are aggravated difficulties of water availability, the Contractor may seek clarifications with the Board to resolve such situations. Should any pipeline fail a test and require subsequent retesting, the Contractor shall supply the water for retesting also.

- (xxii) Unless otherwise specified, the Contractor shall make provisions in the design of pipework for pressure testing. Such provisions shall include filling and draining points, connections for air vacuum release valves, connections for pressure gauges, etc.
 - (xxiii) The Contractor shall provide all pressure gauges, automatic pressure recorder(s), meters, hoses, pumps, stop ends and other temporary supports and equipment needed for carrying out tests and shall carry out all work necessary to test the pipeline. The Contractor shall make all arrangements necessary for conveying the water to the point of filling and such arrangements shall be subject to the approval of the S.O.. The test ends shall be tapped for suitably size test connection.
 - (xxiv) Gauges used for testing pipelines shall either be of the conventional circular type, not less than 200 mm diameter, or shall have a digital indicator. The gauge shall be capable of reading increments of 0.1 metre head. Before any gauge or pressure recorder is used and at other times as the S.O. may direct, the Contractor shall arrange for it to be checked independently and a dated certificate of its accuracy shall be provided. The pressure recorder shall have an 48 hour chart to record the pressure drop continuously.
 - (xxv) The Contractor shall remain responsible for the care of the Works during testing and cleaning of the pipeline.
 - (xxvi) No length of pipeline shall be deemed to have passed its pressure test until the S.O. has issued a certificate to the Contractor to this effect.
 - (xxvii) The Contractor shall take all necessary steps to locate any leakage, supply necessary materials and make necessary repairs or remake any joints, and restore the ground at no extra cost to the Contract until, on retest, the above requirements as to permissible loss or zero leakage are complied with.
- (b) Pressure Testing Plan
- (i) A pressure testing plan shall be submitted 30 days prior to beginning of any pressure testing. Testing plan shall include dates, sections to be tested, test procedure, test equipment, leakage and method of supplying and disposing of water and a certified test report format.
 - (ii) Testing shall not begin until this plan has been approved by the S.O..
- (c) Procedure for Pressure Testing
- (i) All test equipment, including temporary valves, bulkheads or other water control equipment shall be furnished. Do not use materials that would be injurious to the construction or its future function. Remove or isolate items that might be damaged by the pressure testing. Ensure bulkheads and other items are suitably restrained to resist the thrust of test pressures without damage.

- (ii) Before filling for pressure testing is started the Contractor shall re-check pipes and valves for cleanliness and shall re-check the operation of all valves. The 'open' ends of the pipeline (or sections thereof) shall normally be stopped off by blank flanges or cap ends additionally secured where necessary by temporary struts, wedges and thrust blocks. All thrust blocks, diaphragm blocks and valve chambers shall have been completed with backfill placed around them. All pipe straps and other devices intended to prevent the movement of pipes shall have been securely fastened and backfilling of trench between pipe joints shall have been placed to the extent necessary to prevent movement of the pipeline. Where the Contractor has not previously demonstrated to the S.O.'s satisfaction that joints do not leak when subject to air tests, mechanical joints shall be left exposed for inspection during field pressure testing.
- (iii) During pressure testing, joints and connections shall be inspected for leakages. Visible leakages shall be corrected and retested as specified.
- (iv) Fluid used for hydrostatic test for pressure piping shall be cleanwater of such quality to prevent corrosion of materials in piping system.
- (v) Empty pipe of water prior to final cleaning or disinfection. No standing water shall be left in a piping system for more than 1 week to prevent internal pipe corrosion.
- (vi) Hydrostatic testing of exposed piping shall comply with the following additional requirements, as a minimum:
 - 1) Perform testing on installed piping prior to application of insulation;
 - 2) Maximum Filling Velocity: 75 mm per second, applied over full area of pipe;
 - 3) Vent piping during filling. Open vents at high points of piping system or loosen flanges, using at least four bolts, or use equipment vents to purge air pockets; and
 - 4) Maintain hydrostatic test pressure continuously for 60 minutes, minimum, and for such additional time as necessary to conduct examinations for leakage.
- (vii) Hydrostatic testing of buried piping shall comply with the following additional requirement as a minimum:
 - 1) Test after backfilling has been completed on non-restrained piping systems. Test before backfilling on restrained joint piping systems;
 - 2) Expel air from piping system during filling;
 - 3) Apply and maintain specified test pressure with hydraulic force pump. Valve off piping system when test pressure is reached; and
 - 4) Maintain hydrostatic test pressure continuously for 2 hours minimum.
- (d) Pressure Test Report
 - (i) Test Report Documentation shall as a minimum include:
 - 1) Test date;
 - 2) Description and identification of piping tested;

- 3) Test fluid;
 - 4) Test pressure;
 - 5) Remarks, including:
 - Leaks (type, location); and
 - Repair/replacement performed to remedy leakage.
 - 6) Signed by Contractor and S.O. to represent that test has been satisfactorily completed.
- (e) Pressure testing of cement lined pipework
- (i) The pipeline or section of pipeline to be tested shall be filled with clean water and all air shall be removed. Before the test proceeds further, the pipeline shall be allowed to stand full of water for at least 24 hours at the working pressure.
 - (ii) The pressure shall then be raised steadily by pumping in water until the specified test pressure (P) is reached and shall be maintained at this pressure by further pumping as required for a period of at least 24 hours during which time the pipeline shall be inspected for points of concentrated leakage. In the case of pipelines of absorbent materials, this period shall be extended as necessary until the absorption has ceased. The pump shall then be disconnected and no further water shall be permitted to enter the pipeline. The time (T) for the observed pressure head to fall by 10 m shall be recorded. Pumping shall then be resumed and the quantity of water (Q) pumped into the pipeline from calibrated container(s) in order to restore the test pressure (P), shall be recorded. Provided that if the observed pressure head has not fallen by 10 m before an expired period of three hours then pumping shall be resumed at once and the time (T) shall be recorded as being three hours. The rate of loss shall be calculated as being the quantity (Q) divided by the time (T).
 - (iii) Test Pressure (P)
 - 1) The test pressures (P) for the principal pressure pipelines to be laid under this Contract are set out in the Particular Specification.
 - (iv) Test Standard
 - 1) If the rate of loss as calculated above does not exceed the values given below, the pipeline (or section thereof) shall be deemed to have passed the pressure test, provided that there is no evidence of concentrated leakage.
 - Steel pipelines with welded joints
 - The length of main under test shall be deemed to have passed the test if the leakage does not exceed 0.05 litre per mm of internal diameter per 24 hour per kilometre of pipeline.
 - All other pipelines
 - The length of main under test shall be deemed to have passed the test if the leakage does not exceed 0.1 litre per mm of internal diameter per 24 hour per kilometre of pipeline for each 30 m of pressure applied; and

- Where a new pipeline constructed under the Contract is to connect to an operational pipeline the final connection shall be inspected visually under normal operating pressures and there shall be no visible leakage.
- 2) Action following Pressure Test Failure
 - If any pressure pipeline fails to pass the pressure test the Contractor shall locate the faults and shall uncover, repair, retest and reinstate that pressure pipeline as may be necessary until all parts of the pipeline shall have passed the pressure test.
- 3) Washout Pipework
 - A pressure test on washout pipework shall not be required. The performance of washout pipework shall, however, be observed during flushing of the pipelines. Each scour valve in turn shall be opened fully under the normal working head in the pipeline for sufficient time to allow proper flushing of the pipeline and to check the water tightness of the washout pipework. The Contractor shall carry out any necessary remedial work to pipework as directed by the S.O..
- (f) Testing of unlined pipework
 - (i) The pipework other than cement lined pipes shall be tested to at least 1.5 times the maximum operating pressure of the pipeline unless otherwise specified. GRP pipeline shall be tested to 1.5 times the maximum working or the maximum working pressure plus 5 bar whichever is lower or the surge pressure if this is greater. Polyethylene pipeline shall be tested to 1.5 times maximum working pressure for 12 and 16 Bar systems.
 - (ii) Pipes under vacuum shall be tested under vacuum not less than 1.5 times the operating vacuum (gauge).
 - (iii) The pipework shall be allowed to stabilise and shall not show any visible leaks and shall hold the test pressure, once stabilised, for at least one hour, and for such additional time as necessary to conduct examinations for leakage, with no reduction in test pressure with no pumping (to add pressure) and without adding additional water.
 - (iv) Action following Pressure Test Failure
 - 1) If any pressure pipeline fails to pass the pressure test the Contractor shall locate the faults and shall uncover, repair, retest and reinstate that pressure pipeline as may be necessary until all parts of the pipeline shall have passed the pressure test.
- (g) Testing Gravity Pipelines
 - (i) In addition to weld testing, welded steel gravity pipelines shall be tested for infiltration and water tested as described in sections 2) and 3) below. Non welded steel and gravity pipelines of other pipe materials shall in addition be air tested prior to backfilling as described in the sections below:

1) Air Test prior to Backfilling

- Each length of gravity pipeline shall be carefully air tested after they are jointed and before any concreting or backfilling is commenced, other than such as may be necessary for structural stability whilst under test. Air shall be pumped in by suitable means until a pressure of 100 mm head of water is indicated on a manometer tube connected to the system. The pump shall then be disconnected. The pipe shall pass the air test if the loss in air pressure does not exceed 25 mm over a period of five minutes without further pumping, following a period for stabilisation. The pipeline shall be tested in convenient lengths and all necessary tester junctions, expanding rubber plugs, testing equipment etc. shall be provided by the Contractor. The Contractor shall immediately remedy any defect, which may become apparent and re-test to the satisfaction of the S.O..
- With the approval of the S.O. the bedding may then be placed and the trench backfilled in the manner specified.

2) Infiltration Test upon Completion of Backfilling

- Upon completion of backfilling an infiltration test shall be carried out unless otherwise directed by the S.O.. All inlets to the system shall be effectively closed, and any residual flow shall be deemed to be infiltration. The pipeline, including manholes shall be accepted as satisfactory if the infiltration, including infiltration into manholes, does not exceed 1.5 litres per 30 minutes per 300 millimetre of nominal internal pipe diameter per 100 metres of pipe.
- If it appears that any pipe has been broken or any joint has been disturbed during backfilling then the Contractor shall re-open the trench and repair the defect or defects until the pipeline has been retested to the satisfaction of the S.O..
- The Contractor shall ensure that a complete set of testing equipment and all necessary expanding rubber plugs are available at all times at all parts of the Works where pipelaying is in progress in order that the S.O. may carry out a test at any time.

3) Water Testing

- The test pressure for non-pressure pipelines shall be not less than 1.2 m head of water above the pipe soffit or ground water level whichever is the higher at the highest point of the pipe length and not greater than 6 m head of water at the lowest point of the section. Steeply graded pipelines shall be tested in stages in cases where the maximum head, as stated above, would be exceeded if the whole section were tested in one length.
- The pipeline shall be filled with water and a minimum period of 2 hours shall be allowed for absorption, after which water shall be added from a measuring vessel at intervals of 5 minutes, and the quantity required to maintain the original water level noted. The length of pipeline shall be acceptable if the quantity of water added over a 30 minute period is less than 1.5 litres per 300 millimetre of nominal internal pipe diameter per 100 metres of pipe.

- Unless the S.O. shall direct otherwise the Contractor shall be at liberty to retain the water in the pipes after the first test and during the process of backfilling so that the pipe lengths need only to be water filled once for testing purposes.
- (h) Pneumatic Test for pressure Piping
- (i) Do not perform on:
 - 1) PVC or CPVC pipe;
 - 2) Piping larger than 450 mm, except air low pressure and odorous air piping; and
 - 3) Buried and other non-exposed piping.
 - (ii) Fluid used shall be oil-free, dry air.
 - (iii) Pressure testing shall be carried out as follows:
 - 1) Apply preliminary pneumatic test pressure of 170 kPa maximum, or half of the test pressure whichever is greater, to piping system prior to final leak testing, to locate visible leaks. Do not apply greater pressure than specified in the Pipe Schedule. Apply soap bubble mixture to joints and connections, examine for leakage.
 - 2) Correct visible leaks and repeat preliminary test until visible leaks are corrected.
 - 3) Gradually increase pressure in system to half of specified test pressure. Thereafter, increase pressure in steps of approximately one-tenth of specified test pressure until required test pressure is reached.
 - 4) Maintain pneumatic test pressure continuously for minimum of 10 minutes and for such additional time as necessary to conduct soap bubble examination for leakage.
 - 5) Correct visible leakage and retest as specified.
 - (iv) Piping system shall show no visual evidence of leakage.
 - (v) After testing and final cleaning, purge with nitrogen those lines that will carry flammable gases to assure no explosive mixtures will be present in system during filling process.

G4.2.23 Flushing and Sterilization of Pipelines

- (a) Certain pipelines require to be sterilised as set out in the Particular Specification.
- (b) When the final connections have been made and the pipelines have been tested to the satisfaction of the S.O., the pipelines shall be again jointly inspected that they are cleaned before they are thoroughly sterilised and flushed by the Contractor.
- (c) The water required to carry out sterilisation of pipelines shall be deemed to have been included under the Contractor's scope of supply. Where situations are encountered in which special considerations render water unavailable or there are aggravated difficulties of water availability, the Contractor may seek clarifications with the Board to resolve such situations.

- (d) The Contractor shall supply all equipment necessary to sterilise the pipelines including meters, hoses, pumps, dosing system and other equipment needed for carrying out sterilisation and shall carry out all work necessary to sterilise the pipeline. The water used for sterilisation shall be potable. The Contractor shall make all arrangements necessary for conveying the water to the point of filling and such arrangements shall be subject to the approval of the S.O..
- (e) The operation of all valves including scour and air valves shall be checked by the Contractor and any necessary adjustments shall be made to ensure correct operation.
- (f) The Contractor's sterilisation procedure for pipelines shall strictly adhere to the Continuous Feed Method as set out in Section 5.2 of AWWA Standard C651-92, 'Disinfecting water mains'. The Contractor shall have to adopt a dosing system to sterilise pipelines effectively. The dosing system shall be provided with a set-up of temporary equipment, chemical mixing tank mixer, mixing tank, dosing pump and other necessary accessories for the solution to be injected into the pipelines when potable water will be introduced gradually. The piping main shall be deemed to have been sterilised if the samples of water taken from various tappings on the main conform to the S.O.'s chemical, physical and bacteriological requirements.
- (g) After the main has been sterilised to the satisfaction of the S.O. it shall be flushed with clean water.
- (h) Pipelines which do not require to be sterilised shall nevertheless be jointly inspected that they are clean upon completion of final connections and testing. Such pipelines shall then be thoroughly flushed by the Contractor.

G4.2.24 Design, Installation and Building in of Pipework

- (a) All pipework provided under the Contract shall be erected and fixed in position by the Contractor. The design of pipework systems shall be take account of thrusts developed within the systems. Wherever possible, thrusts shall be resisted within the pipework system by means of flanged or welded joints or tied couplings, or shall be transmitted direct to the structure by building in anchor flanges into the structure. Couplings or flange adaptors shall be provided where necessary to enable valves and other pipework items to be removed and replaced without dismantling the complete pipework system. All pipes passing through water retaining structures, or pipes entering or emerging from structures below finished ground level shall have puddle flanges.
- (b) Complete sections of pipework, including the items to be built into structures, shall be erected and checked by the Contractor for line and level before any items are built in. Casting in of individual pipes in isolation of the rest of the system shall normally not be permitted, and any proposals for special applications or situations in this respect shall be subject to the approval of the S.O..
- (c) The Contractor shall erect and fix securely in position all pipework which is to be built in. Any supports shall be clear of the openings into which pipes are to be built, to allow access for installation of formwork and placing concrete.

- (d) After the alignment and level of the pipework is complete, subject to the approval of the S.O., the operation of building in shall commence. Formwork shall be erected and concrete placed. The Contractor shall re-check the pipework alignment after erection of formwork and before placing concrete starts, to ensure that it has not moved.

G4.2.25 Work inside Pipelines

- (a) The Contractor shall provide, operate and maintain adequate systems of access, lighting and ventilation to any part of a pipeline where work is in progress inside the pipes.

G4.2.26 Dangers of Floatation

- (a) Although floatation of pipes in completely refilled trenches or of the completed works will not occur, the Contractor's attention is particularly drawn to the danger of floatation of pipes laid in trenches left unfilled and allowed to be flooded with water in the partially completed works.
- (b) The Contractor shall, immediately after pipes have been laid, jointed and approved by the S.O, refill and compact excavated material as specified to at least two thirds of the length of the pipe trench leaving only sufficient space unfilled to allow for the completion of the external joint. Notwithstanding this, the Contractor shall make all necessary provision to avoid the floatation of any pipe during the construction stage.
- (c) Any damage caused by floatation of pipes, specials or fittings in unfilled trenches shall be made good to the satisfaction of the S.O., at no extra cost.

G4.2.27 Pipelines in the Same Trench

- (a) Where two or more pipelines are detailed as being laid in the same trench the pipes shall be laid so that there is a minimum distance of 300 mm between the barrels of the pipes, measured in plan at mid-barrel height, unless a greater distance is shown on the 3D model or Drawings or is required to enable jointing to be properly completed. The invert levels of the pipelines shall be the same at any cross section, unless otherwise required.
- (b) The larger of the pipelines shall be laid in straight lines between bends as shown on the 3D model or Drawings. Where it is necessary to increase the distance between the pipelines from that specified above to allow the construction of air valve or washout chambers or for any other reason, the deviation of a pipe from the line of the preceding pipe shall be made by deflecting as many flexible joints as necessary to limit the amount of deflection at each joint to not more than half the allowable deflection.

G4.2.28 Anchor and Thrust Blocks

- (a) Concrete thrust and anchor blocks in accordance with the details shown on the 3D model or Drawings shall be constructed at all tees, bends, tapers, valves and hydrants for the anchorage of the pipeline. Thrust forces' calculations and thrust blocks design drawings shall be submitted to the Board for approval. Concrete shall be of the grade shown on the 3D model, Drawings or as specified in the Particular Specification.

- (b) Unless otherwise detailed on the 3D model or Drawings, thrust blocks shall be constructed with the bottom and thrust side surfaces bearing against undisturbed ground.
- (c) Shear keys shall be formed on the upper surface of concrete blinding for horizontal thrust blocks as shown on the 3D model or Drawings.
- (d) Where pipes are laid at gradient steeper than 1 in 12, diaphragm blocks in concrete shall be provided at the middle of pipes as follows:

Gradient exceeding	Diaphragm blocks
1 in 12	Every third pipe
1 in 8	Every second pipe
1 in 3	Every pipe

- (e) The dimensions on the 3D model or Drawings shown may, however, be amended by the S.O. after examination of the ground prior to concreting, having regard to the likely thrust developed under test pressure and the actual nature of the ground. Additional diaphragm blocks may also be needed. For horizontal bends in cross fall ground, care shall be taken to ensure that additional cover over the pipes is allowed in order that the same passive resistance may be developed as would be the case in level ground.

G4.2.29 Pipelines Built Into Structures

- (a) Any pipeline which is built into a chamber, manhole or other structure, including a thrust block, shall be provided with two flexible joints outside the face of the structure. Unless otherwise detailed on the 3D model or Drawings, the first joint shall be not more than one pipe diameter or 1000 mm, whichever is the smaller, from the outside face, and the length of the short pipe between the flexible joints shall be as follows :

Pipe Nominal Internal Diameter (mm)	Rocker Pipe Length (m)
150 - 600	0.6
Over 600 - 750	1.0
Over 750	1.2

- (b) The flexible joints shall be either flexible spigot and socket joints or detachable flexible couplings.
- (c) Pressure pipes passing through the walls of the valve chambers or other structures shall be ductile iron or steel to be provided, where detailed on the 3D model or Drawings, with anchorage flanges designed to transmit full end thrust with closed valve under test into the structures wall. Boxouts, if used, shall be designed to fulfil the above requirements. Drawings showing the method of construction using boxouts shall be submitted for approval by the S.O..

- (d) Where no anchorage flanges are detailed for pipes passing through the walls of structures the pipe shall be provided with a puddle flange integral with the pipe wall or bolted on to resist movement of water along the pipe to concrete interface. Such puddle flanges shall protrude at least 50 mm from the pipe barrel.
- (e) In the case of clayware gravity pipelines the short length of pipe built into the wall shall be of the same material and shall be surrounded in concrete as detailed on the 3D model or Drawings for manholes. Where uPVC pipes are shown on the 3D model or Drawings as being built into walls or surrounded with concrete they shall be protected with wrapping material, with a 55 % lap, to a point 50 mm outside the face of the wall or concrete surround.
- (f) Where the pipeline is above ground level the short length of pipe providing flexibility shall be self-supporting.

G4.2.30 Installation of Pipelines other than Buried Pipelines

- (a) The Contractor shall take due care to support adequately the pipework during installation until permanent supports and anchorages as detailed on the 3D model or Drawings or as approved by the S.O. have been completed. The Contractor shall ensure that no excessive loads or stresses are imposed upon the plant, pipework or structure during installation.
- (b) Pipework systems shall be complete and checked for correct position and alignment with adjacent plant and structures immediately before and after being embedded in concrete. Brackets and fixings shall be painted with two coats of bituminous paint to BS 3416 (excluding Clause 10).

G4.2.31 Connections between Different Pipe Materials

- (a) Where different types or classes of pipe material are jointed together, the Contractor shall supply special fittings or stepped couplings designed and manufactured to suit the ends of pipes to be jointed. Stepped couplings shall comply where applicable with the requirements for flexible couplings and flange adaptors as specified hereafter.

G4.2.32 Connecting to Existing Pipelines

- (a) Where the 3D model or drawings indicate that a connection of any kind is to be made into an existing pipeline, or whenever directed by the S.O. to make such a connection, the Contractor shall verify before starting work on the connection, if necessary by excavating trial pits, that the materials to be furnished under the Contract will be suitable for making the connection.
- (b) Before making a connection into an existing pressure pipeline, the Contractor shall liaise with the authority responsible for the operation of that pipeline and shall locate the valves governing the flow to the section of pipeline in question. The Contractor shall prepare a detailed programme for the completion of the connection, clearly indicating the period for which it will be necessary for the existing pipeline to be taken out of service and any measure that will be taken to prevent contamination of the existing pipeline. The programme shall also indicate the method and proposed standby plant to be made available. The period of planned shutdown shall be kept to the minimum practicable in the circumstances. The Contractor's programme shall be subject to the approval of the S.O. and of the operating authority.

- (c) Work on the connection shall be undertaken in accordance with the approved programme. Unless otherwise agreed in writing, the operation of valves necessary to isolate the section of the existing pipeline and subsequently to restore it to service shall only be undertaken by staffs of the operating authority. If so instructed by the S.O., the Contractor shall assist the authority in carrying out all activities necessary to facilitate the efficient completion of work on the connection.
- (d) The existing pipeline to which the connection has been made as well as the new pipeline shall then be tested for leakage. Any existing pipe, fitting or joint leaking due to damage occurring during the connection operation shall be removed and replaced by the Contractor at no extra cost to the Board. The test shall be repeated until a satisfactory test is obtained. Any additional cleaning and disinfection procedures required in respect of the connection shall be performed by the Contractor at no extra cost to the Board.
- (e) Where testing of the existing pipeline is not practicable and where directed by the S.O., the new pipeline shall be tested up to the point of connection. After testing, the connection shall be made up and left exposed for visual inspection. There should be no visible leakage detected under normal operating pressure for a period of 24 hours or in accordance with the Particular Specification for the specific applications.

G4.2.33 Plugging of Dead Ends

- (a) The dead ends of pipes and fittings shall be plugged as indicated on the 3D model or Drawings or as directed by the S.O.. Anchors shall be provided as necessary.

G4.2.34 Cables Sharing Pipeline Trenches

- (a) Where cables are detailed on the 3D model or Drawings to be laid in pipeline trenches, the pipes shall be installed and backfilled with the cables placed to 150 mm above the crown of the pipe. The cable shall then be laid as shown on the 3D model or Drawings before the remainder of the trench is backfilled.

G4.2.35 Marker Tape for Buried Services

- (a) The marker tape laid above buried services shall be 500 mm wide and 0.1 mm nominal thickness coloured polyethylene, boldly printed throughout its entire length with the word "CAUTION" and words to identify the particular service, in accordance with the following table:

Types of service	Colour of tape	Service identification wording
Potable water	Blue	WATER MAIN BELOW
Wastewater, effluent and sludge of all types	Green	WASTE PIPE BELOW
Fuel oil or gas pipe	Orange	FUEL MAIN BELOW or GAS MAIN BELOW
Electricity, earthing	Yellow	ELECTRIC CABLE BELOW
Instrumentation cable Telephone cable Telemetry cable	Red	CONTROL CABLE BELOW

G4.2.36 Route Marker Posts

- (a) The Contractor shall provide and fix precast concrete indicator posts at every 50 m and change of pipe direction, details of which are shown on the 3D model or Drawings on the centreline of the pipes at bends, valves, washout, air valves and other points as directed by the S.O.. Name and number plates fabricated from 16 gauge aluminium sheets with embossed lettering shall be fixed to the concrete post with stainless steel screw and fibre glass washers. The plates shall indicate the type of services, direction of flow, diameter and depth of the main or cable below ground level.
- (b) The marker posts shall be not less than 300 mm x 150 mm and shall be designed with suitable footings and the proposal shall be submitted to S.O. for approval.
- (c) The marker posts shall set up as soon as practicable after trench excavations have been refilled but the Contractor shall provide any temporary marking required and shall be responsible for correct information being given on the indicator plates.
- (d) The post shall be set in concrete firmly into the ground to the depths shown on the 3D model or Drawings with the backfilling well rammed in and around.

END OF SECTION